Event Horizon Fund: Comprehensive Deep Dive Analysis & Optimization Report

Date: May 10, 2026 **Cycle:** V9.1 (5 follow-up reconciliation changes adopted; supersedes V9.0) **Scope:** Full portfolio re-evaluation against Q1 2026 earnings, regulatory developments, and the migration of binding bottlenecks from compute/packaging to **power and optics**. **Inputs synthesized:** prior V8.3 analysis (March 31, 2026); GPT5.5 Pro Research (May 5, 2026); Gemini Pro Research (May 5, 2026); follow-up reconciliation by Gemini + GPT5.5 (May 10, 2026); fresh stock-by-stock research with timestamps; AI-2027, METR, Epoch, and Aschenbrenner frameworks.

> ### ⚠️ Active Allocation Version: V9.1 (supersedes V9.0)
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> The V9.1 allocation is the **active, canonical portfolio** as of May 10, 2026. It reflects 5 high-conviction reconciliation changes adopted after independent review by Gemini Pro Research and GPT5.5 Pro Research.
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> **V9.1 changes vs V9.0:** CEG 11%→9%, DELL 4%→2%, VST 4%→6%, GEV 4%→5%, PWR 3%→4%. All other 18 holdings unchanged. Total = 100.0%.
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> V9.0 references throughout this document remain for historical/audit context but are **no longer the active allocation**. For full V9.0 → V9.1 change rationale, see **Part XIII: V9.0 → V9.1 Reconciliation (Follow-up)**.

Part I: Executive Verdict

The Thesis Has Strengthened — But the Bottleneck Has Migrated

After re-validating every position against Q1 2026 results, regulatory developments, and the latest GPT5.5 + Gemini research inputs, **the core thesis is more strongly validated than at the V8.3 cycle, not less.** Hyperscaler 2026 capex has been revised UP from prior estimates of $660–690B to **$725–830B** (TrendForce / public earnings). Microsoft alone now guides $190B (+130% YoY). Power demand at the FERC-approved cap. Optics screaming through 90% growth. Every "must-be-true" threshold continues to be met or exceeded.

But the location of the binding constraint has shifted. The pure-compute and advanced-packaging chokepoints (TSMC CoWoS, HBM, Blackwell) are still expanding aggressively. The new binding constraints are:

1. **Front-of-meter and behind-the-meter power delivery** — PJM 2026/27 capacity auction cleared at the FERC ceiling of $329.17/MW-day; data centers drove 40% of the $16.4B capacity cost; co-located load is now formally defined and getting structured interconnection paths.
2. **Optical fabric inside hyperscale clusters** — 800G ramping aggressively, 1.6T entering production in 2027, supply imbalance >30% in EML/InP lasers; Nvidia took a $2B equity position in Coherent.
3. **On-site distributed generation** — the Oracle/Bloom Energy 2.8 GW Project Jupiter agreement (announced April 27, 2026) validates fuel cells as primary baseload for gigawatt-scale AI campuses, bypassing 5–7 year interconnection queues.

The portfolio must evolve to capture these newly binding bottlenecks.

Key Conclusion on Current Allocation

The V8.3 allocation is **roughly 80–85% optimal** in the new environment, slightly down from 85–90% optimal at the time it was set. The decline is not because old positions failed — they nearly all crushed Q1 — but because **two new investable bottlenecks have emerged** that the V8.3 portfolio does not address. Those gaps are: - **Grid-bypass power (Bloom Energy, BE)** - **Optical fabric / co-packaged optics (Coherent, COHR)**

The recommended V9.0 changes are evolutionary, not revolutionary. They add two new core positions, modestly trim valuation-extended winners, increase a few names where Q1 earnings raised conviction, and keep the overall sleeve structure intact.

V9.1 Action Summary (Active — 5 follow-up changes vs V9.0)

After independent review by Gemini Pro Research and GPT5.5 Pro Research (see `Analysis 2026-05-10/Follow Up/Followup Cursor Results 5-10-2026.md`), 5 high-conviction reconciliation changes were adopted:

1. **Trim Constellation (CEG) from 11% → 9%** — both followups agree single-name/event risk too high; FERC co-location friction
2. **Trim Dell (DELL) from 4% → 2%** — both followups agree integrator economics, not a bottleneck owner
3. **Increase Vistra (VST) from 4% → 6%** (funded by CEG) — both followups agree faster growth + IG upgrade + oil-shock resilience
4. **Increase GE Vernova (GEV) from 4% → 5%** (funded by DELL) — both followups agree $163B backlog + transformer monopoly
5. **Increase Quanta (PWR) from 3% → 4%** (funded by DELL) — record $48.5B backlog + cleanest grid-buildout toll road

Net effect: Power chain (CEG+VST+PWR+GEV+BE+CCJ) sharpens from 27% → **29%**; Rack-Scale (VRT+DELL) trims from 10% → **8%**. All other 18 holdings unchanged.

V9.1 Total checksum: 100.0% (verified)

Rejected from followups: add CORZ/IREN (Moonshot/LEAPS targets — deferred), swap COHR→LITE (COHR has stronger Nvidia $2B validation), raise BE to 6% (Gemini's chase rejected, current 3% balanced), convert INTC to LEAPS (no LEAPS per user rule), add ETN (GEV already covers electrical bottleneck), add CDNS (comparable to SNPS, not superior).

V9.0 Action Summary (Historical — 8 changes from V8.3, superseded by V9.1)

1. **Add Bloom Energy (BE) at 3%** — fund the grid-bypass thesis (Oracle 2.8 GW deal is a paradigm-validating event)
2. **Add Coherent (COHR) at 2%** — fund the optical fabric thesis (Nvidia $2B equity stake confirms the bottleneck)
3. **Increase Vistra (VST) from 3% → 4%** — Q1 crushed (rev +43%, EPS $2.87 vs $2.21 est, Investment-Grade upgrade by Fitch)
4. **Increase GE Vernova (GEV) from 3% → 4%** — backlog $163B (+$13B sequentially), gas turbine backlog now 100 GW (up from 83 GW)
5. **Trim Constellation (CEG) from 12% → 11%** — modest trim for concentration; Q1 guide slightly below consensus; VST is the higher-growth complement
6. **Trim TSM from 12% → 11%, VRT from 8% → 6%, DELL from 5% → 4%, ANET from 5% → 4%, INTC from 4% → 3%** — modest valuation/concentration trims; all positions still core
7. **Hold every other position unchanged** — CRWD, PLTR, MU, ASML, AMAT, NVDA, AVGO, SNPS, AVAV, KTOS, MRNA, MP, PWR, CCJ
8. **Hold Moonshot at 0%** — LEAPS strategies still pending; CORZ/IREN/CRWV deferred to that sleeve

V9.0 Total checksum: 100.0% (verified — historical baseline before V9.1 adjustments)

Part II: What Changed Since the V8.3 Cycle (March 31 → May 10, 2026)

This is the most important section for cycle-to-cycle continuity. Only **new evidence** is listed here.

A. Earnings Surprises (Q1 2026 Reality vs. V8.3 Expectations)

Metric	V8.3 Expectation (Mar 31)	Q1 2026 Reality	Delta
Hyperscaler 2026 capex	$660–690B	**$725–830B (raised)**	**Far exceeded**
Microsoft 2026 capex	implicit ~$140B	**$190B (+130% YoY)**	Far exceeded
Meta 2026 capex	implicit ~$110B	**$125–145B**	Exceeded
Amazon 2026 capex	implicit ~$200B	**>$230B**	Exceeded
TSMC 2026 revenue growth	mid-20s%	**>30% (raised)**	Exceeded

Metric	V8.3 Expectation (Mar 31)	Q1 2026 Reality	Delta
ASML 2026 revenue	~€33B	**€36–40B (raised)**	Exceeded
GE Vernova backlog	~$150B	**$163B (+$13B QoQ)**	Exceeded
GEV gas turbine backlog	83 GW	**100 GW (target 110 GW EOY)**	Exceeded
Vertiv 2026 revenue	~$11B	**$13.5–14B (raised)**	Exceeded
Quanta backlog	~$36B (Sept 2025)	**$48.5B record**	Exceeded
Cameco Q1 EBITDA	~$420M	**$509M**	Exceeded
Dell FY27 AI server	$50B (guide)	**$50B (re-confirmed)**	On target
Dell Q4 AI revenue	strong	**$9B (+342% YoY)**	Exceeded
Palantir 2026 revenue	~$5.7B (guide)	**$7.65B (raised, +71% YoY)**	Far exceeded
Palantir Q1 growth	~60% expected	**+85% (fastest since IPO)**	Far exceeded
CrowdStrike net new ARR (Q4)	strong	**+47% YoY ($331M record)**	Exceeded
Vistra Q1 EPS	$2.21 consensus	**$2.87 (+30% beat)**	Far exceeded
Vistra credit rating	BB+	**Investment Grade (Fitch upgrade)**	Achieved
Lumentum Q3 revenue	~$700M	**$808M (+90% YoY)**	Exceeded
Coherent Q3 revenue	~$1.7B	**$1.81B + Nvidia $2B equity**	Exceeded
Bloom Energy Q1 revenue	not on radar	**$751M (+130% YoY)**	New paradigm
CoreWeave Q1 revenue	not on radar	**$2.08B (+91%); $99.4B backlog**	New scale

B. Regulatory Developments

- **PJM 2026/27 BRA cleared at FERC cap of $329.17/MW-day** (Dec 2025 results, fully digested by April 2026). Data centers attributable for ~40% of the $16.4B cost. This is structural validation of the power thesis.
- **FERC April 16, 2026 co-location ruling** — partially accepted, partially rejected PJM's compliance filing. Crucially, FERC **reaffirmed the Commission-approved Co-Located Load definition** AND opened pathways for "reduced-capacity interconnection, provisional service, and surplus service options." Net: the regulatory environment for nuclear-to-datacenter deals is **clearer and slightly more accommodating**, not "severely friction-ridden" as one input report suggested.
- **DOE approved $1B loan guarantee** for Constellation's Crane Clean Energy Center restart (TMI Unit 1).
- **Federal nuclear support intact** — Nuclear PTC up to $15/MWh continues to support CEG/VST nuclear economics.

C. Sector Inflection Events

- **Oracle / Bloom Energy 2.8 GW deal** (April 27, 2026) — Project Jupiter in Doña Ana County, NM, will be powered entirely by **up to 2.45 GW of Bloom Energy fuel cells**. This is the largest on-site fuel cell deployment ever announced and **paradigm-changing** for AI datacenter power architecture.

- **Nvidia $2B equity investment in Coherent + multi-year supply agreement** announced with Q3 results — anchor-customer validation for the optical fabric thesis.
- **CoreWeave Q1: $99.4B backlog**, including a **$21B Meta deal** signed in March; expanded contracted power to 3.5 GW, targeting 8 GW by 2030.
- **IREN secures $9.7B Microsoft AI cloud contract** (5-year, 20% prepayment); $5.8B Dell GPU/equipment supply deal; stock +613% YoY.
- **Core Scientific delivering 243 MW to CoreWeave at 80–85% gross margins** (raised from 75–80%); on track for 590 MW by early 2027.
- **Eaton (ETN) data center orders +240% YoY**; closed $11B in M&A (Boyd Thermal $9.55B + Ultra PCS $1.53B) targeting datacenter and aerospace.

D. Stock Recovery Since March Lows

- CEG: $279 (Mar 31) → ~$307 (May 1) — recovered **+10%**, but still below October 2025 ATH of $413.
- VST: ~$150 (Mar 31) → ~$148 (May 8) — flat after a strong Q1 beat (consolidating).
- TSM, ASML, NVDA, MU, GEV, PWR, VRT — all participated in the broader market recovery; most are at or near 52-week highs.

E. Macro Backdrop Has Inverted

- **March 31, 2026:** S&P -9% from highs, VIX 31, RSI deeply oversold, Iran April 6 deadline imminent → "buy the blood" environment.
- **May 10, 2026:** S&P at record highs (~7,398), VIX ~17, Mag 7 +14.88% in April, Q1 earnings growth +27.1%, fwd P/E 20.9 (elevated). Iran negotiations have collapsed; oil >$114/bbl. **Different environment — discipline, not urgency.**

Part III: Background Framework Validation (Refreshed)

A. AI-2027 Scenario (as of May 2026)

The AI-2027 scenario continues to track reality with uncanny precision:

- **Coding automation phase:** confirmed. Frontier coding models (GPT-5.5, Claude Sonnet 4.5, Gemini 2.5 Pro) now reliably handle multi-hour software tasks. AI-assisted code generation is now mainstream in major AI labs and big-tech engineering organizations.
- **AI R&D progress multiplier:** OpenAI, Anthropic, and Google publicly report ~1.5–2x R&D acceleration from internal AI assistants. Trajectory is roughly on AI-2027's predicted curve.
- **Compute demand collision with physical constraints: violently confirmed.** The PJM cap auction, 100 GW gas turbine backlog at GEV, and Bloom's grid-bypass deal are real-world manifestations of the scenario's Tier-3 bottleneck.

- **U.S.–China dynamics:** Export controls have tightened further; China's compute capacity remains constrained relative to U.S. hyperscalers. The "sovereign premium" continues to be priced into INTC, MP, and domestic-foundry-adjacent assets.

Verdict: AI-2027's "Mid-2026: Agent Workflows" phase is unfolding largely as predicted.

B. METR Long-Horizon Tasks (as of May 2026)

METR's published trend (50% reliability task horizon doubling every ~4–7 months) is holding. By mid-2026, frontier agents reliably complete tasks taking humans several hours; there are early signs of multi-day autonomous task completion in software engineering. **Investment implication unchanged:** demand for compute, power, memory, and cooling continues to accelerate.

C. Epoch AI Trends (as of May 2026)

- Training compute still growing ~5x/year for frontier models.
- Algorithmic efficiency improving ~3x/year.
- LLM inference prices falling ~40x/year (now under $0.20 per million output tokens for top-tier models).
- Hardware FLOP/s per dollar improving ~1.37x/year.

The Jevons Paradox is fully in effect. Cheaper inference is creating MORE total demand for compute, which is precisely why hyperscaler capex is doubling, not falling.

D. Aschenbrenner / Situational Awareness LP (Fund Positioning, May 2026)

The Situational Awareness LP fund (estimated ~$9B AUM, top-10 concentration ~86%) provides a useful tactical benchmark. Its largest positions per the latest Gemini-sourced filings:

Aschenbrenner Position	Approx. Weight	Implication
Bloom Energy (BE)	16–22%	Validates grid-bypass thesis; reinforces V9.0's BE addition
CoreWeave (CRWV, equity + options)	~22%	Compute-landlord thesis; defer to LEAPS sleeve
Intel (INTC call options)	~14%	Sovereign foundry geopolitical hedge; we hold equity not options
Lumentum (LITE)	9–12%	Optical fabric; we use Coherent (COHR) instead — stronger validation via Nvidia $2B stake
Core Scientific (CORZ)	~11%	Compute landlord; defer to LEAPS
Iris Energy (IREN)	~8%	Compute landlord; defer to LEAPS

Adoption verdict: We adopt the **directional read** (grid bypass + optical) but with risk-adjusted vehicle choices (BE equity, not high-leverage option structures; COHR, not LITE; CORZ/IREN/CRWV deferred to LEAPS).

Part IV: Sleeve-by-Sleeve Review

Each sleeve is reviewed against the V8.3 baseline, Q1 2026 evidence, and at least two credible alternatives (per the prompt template).

1. Energy & Power Generation — 18% (CEG 11%, VST 4%, PWR 3%)

Status: REBALANCED — CEG -1%, VST +1%, PWR unchanged.

The single most important sleeve in the portfolio. Power continues to be the binding bottleneck, validated by the PJM cap auction ($329.17/MW-day) and the 5,100 MW of YoY peak load growth attributable to data centers.

Constellation Energy (CEG) — Recommend 11% (-1% from V8.3)

Q1 2026 results due May 7-11; March 31 guidance came in slightly below the $11.60 consensus at $11.00–12.00 EPS midpoint. The 22 GW nuclear fleet remains the irreplaceable strategic asset; the Microsoft TMI restart (835 MW) and Meta Clinton (1.1 GW) PPAs remain on track; DOE $1B loan guarantee for Crane is now approved. The April 16 FERC ruling is **less restrictive than the Gemini input report suggested** — it reaffirmed the Commission's Co-Located Load definition and opened reduced-capacity interconnection paths.

That said, at 12% CEG was the largest single-name position, and its 2026 EPS guide came in below consensus. A modest 1% trim reduces concentration risk without abandoning the position. CEG is still by far the largest power-generation holding.

Vistra Corp (VST) — Recommend 4% (+1% from V8.3)

Vistra's Q1 2026 was the standout quarter in the entire portfolio. Revenue $5.64B (+43% YoY, beat $5.45B est); Adjusted EBITDA $1.49B (+20%); EPS $2.87 vs $2.21 consensus (+30% beat); net income swung from -$268M to +$1.03B. **Most importantly: Fitch upgraded VST to Investment Grade** (joining S&P) — a structural cost-of-capital and bidding-pool improvement. Reaffirmed 2026 EBITDA guidance of $6.8–7.6B. Meta nuclear PPA (~2.6 GW) and Cogentrix gas acquisition (5.5 GW) both **excluded from current guidance**.

VST trades at $147 (May 8) vs analyst consensus target ~$237 — meaningful upside if the gas + nuclear blended model continues to attract hyperscaler counterparty demand. Increasing from 3% → 4% is justified by the Q1 beat, the credit upgrade, and 66% earnings growth vs CEG's ~24%.

Quanta Services (PWR) — Recommend 3% (unchanged)

Q1 2026 was excellent: revenue $7.87B (+26.3%), record backlog $48.5B, raised 2026 guide to $34.7–35.2B revenue and $13.55–14.25 EPS. **Tech & load center revenue growth expectation raised from 70% to 110%.** The company is investing $500–700M to **double its transformer manufacturing capacity** — directly addressing the 128-week transformer lead time crisis. Holding at 3%; PWR is not a concentration risk and the backlog visibility is exceptional.

Alternatives evaluated:

Alternative	Theme	Better/Comparable/Worse	Decision
Talen Energy (TLN)	Pure nuclear (Susquehanna + AWS PPA)	**Worse** (single-asset concentration; 2.2 GW nuclear vs CEG's 22 GW)	Not added
NRG Energy	Integrated power + retail	Worse for thesis (consumer retail dilutes AI exposure)	Not added
NextEra (NEE)	Renewables + regulated utility	Worse (renewables are supplement, not baseload for 24/7 AI)	Not added
XLU ETF	Broad utilities	Worse (dilutes thesis; Continuity Fund handles broad market)	Not added

Sleeve total unchanged at 18%.

2. Foundry & Advanced Packaging — 11% (TSM)

Status: MODEST TRIM — TSM 12% → 11%.

TSMC continues to be the central chokepoint for all advanced AI silicon. Q1 2026: revenue $35.9B (+35–40% YoY), gross margin 66.2%, operating margin 58.1%, net income +58% YoY. HPC = 61% of revenue, +20% sequentially. **Management raised 2026 revenue growth guidance to >30% in USD terms.** Q2 guide $39.0–40.2B at 65.5–67.5% gross margin. Capex at high end of $52–56B; supply constraints expected to persist into 2027.

The thesis is intact and arguably stronger. The trim from 12% → 11% is **purely a concentration management decision**, not a thesis change. TSM remains the single largest foundry/packaging exposure and benefits directly from every dollar of hyperscaler capex flowing through Nvidia, AMD, AVGO, AAPL, and others.

Alternatives evaluated: - Intel (INTC): captured separately (see Domestic Foundry sleeve); not a substitute for TSM but a hedge. - Samsung Foundry: not investable in U.S. equity markets cleanly.

3. WFE / Semi Capital Equipment — 12% (ASML 8%, AMAT 4%)

Status: UNCHANGED.

ASML Q1 2026: net sales €8.8B, gross margin 53.0% (high end of guidance), net income €2.8B, EPS €7.15. EUV represented 66% of system sales (up from 48% prior quarter). **Raised full-year 2026 revenue guidance to €36–40B** (from €30–35B previously). Q2 guide €8.4–9.0B. Two High-NA EUV systems shipped in Q1.

AMAT continues as the diversified U.S.-centric WFE play. CHIPS Act fab construction in Arizona, Ohio, Texas, and Idaho all require AMAT equipment. The advanced packaging push (>$3B target) is on track.

The "geopolitical multiplier" is fully in effect. Parallel fab construction in U.S., Europe, Japan, and South Korea all require duplicate WFE equipment from the same oligopoly. Holding at 12% combined.

Alternatives evaluated: - Lam Research (LRCX) and KLA (KLAC): strong oligopolists, but adding them would over-diversify the WFE sleeve. ASML's EUV monopoly + AMAT's breadth is the cleaner expression.

4. Accelerators & Custom Silicon — 9% (NVDA 6%, AVGO 3%)

Status: UNCHANGED.

Nvidia: Q4 FY26 record $68.1B revenue (+73% YoY), data center $62.3B (+75% YoY); FY26 full year $215.9B (+65%). **Q1 FY27 outlook: $78B (±2%)** — implying continued ~50%+ growth. Rubin platform shipping to customers in H2 FY27 with **10x lower inference cost** than Blackwell.

Broadcom: Q1 2026 AI chip revenue $8.4B (+106% YoY); total revenue $19.31B (+29%). CEO guides AI chip revenue **>>>$100B in 2027** (from chips alone). Six confirmed custom-ASIC customers including Google (TPU next-gen), Meta (multi-GW XPU), Anthropic (3 GW), OpenAI (>1 GW). Q2 guide: $22B total / $10.2B AI.

Both names are doing what we asked them to do: NVDA owns training; AVGO owns the custom-ASIC workaround. The 6%/3% split remains correctly calibrated; the deliberate underweight relative to upstream bottlenecks (TSM, ASML, MU, power) remains correct because pricing power is more concentrated upstream.

Alternatives evaluated: - AMD: still credible #2 GPU but doesn't own a unique bottleneck (its TSMC capacity is already captured via TSM). Pass. - Marvell (MRVL): solid custom-silicon adjunct but smaller scale and weaker hyperscaler relationships than AVGO. Pass.

5. Memory (HBM) — 6% (MU)

Status: UNCHANGED.

Micron Q3 FY26 guidance: **$33.5B revenue (~40% sequential), 81% gross margin, $19.15 non-GAAP EPS.** This is a once-in-a-generation memory cycle; HBM gross margins running near software-company levels. Calendar 2026 HBM supply (including HBM4) **fully sold out on price and volume**. HBM TAM: $35B (2025) → ~$100B by 2028 at ~40% CAGR.

The structural transformation from cyclical commodity to AI strategic supplier is now fully priced in but continues to deliver. 6% remains the right size — meaningful but not over-concentrated in a name that could still face cyclicality if commodity DRAM oversupplies.

6. Rack-Scale Integration & Cooling — 10% (VRT 6%, DELL 4%)

Status: REBALANCED — VRT 8% → 6%, DELL 5% → 4%.

Vertiv (VRT) — Recommend 6% (-2%). Q1 was excellent: revenue $2.65B (+30%), Americas +44%, full-year guidance raised to $13.5–14.0B (+34% growth) and $6.30–6.40 EPS (+51%). Backlog supports the thesis. However, the stock now trades around 87–88x trailing earnings and ~48x forward — full valuation. The trim to 6% maintains meaningful exposure while reducing single-name valuation risk. VRT remains the dominant liquid-cooling and rack-power-management vendor.

Dell (DELL) — Recommend 4% (-1%). Q4 FY26 was very strong: full-year revenue $113.5B (+19%), AI server revenue $25.2B (FY26), Q4 AI revenue $9B (+342% YoY); enters FY27 with a $43B AI backlog and FY27 guide of $138–142B (+23% midpoint, AI ~$50B +103%). 20% dividend increase + $10B buyback authorization. **Excellent execution — but DELL is a systems integrator, not a bottleneck owner.** Competition with HPE and tier-2 ODMs persists. The trim from 5% → 4% redirects capital to harder bottlenecks (BE, COHR, GEV).

Alternatives evaluated: - HPE: direct competitor; same business model issues; pass. - SMCI: governance concerns; explicitly excluded. - Eaton (ETN): see "Cross-Sleeve Candidates" — more diversified industrial, captured indirectly via PWR/GEV.

7. EDA Software — 5% (SNPS)

Status: UNCHANGED.

Synopsys Q1 FY26 revenue $2.409B (+66% YoY incl. Ansys), 42.1% non-GAAP operating margin. Design Automation (which now includes Ansys) at $2B (+96% YoY). Reiterated FY26 guide $9.61B; raised FY26 EPS guide to $14.38–14.46.

The custom-silicon multiplier continues to expand the EDA TAM as every hyperscaler designs its own chips. The Synopsys + Ansys combination creates joint solutions in 1H 2026 with monetization in FY27.

Alternatives evaluated: - Cadence (CDNS): the other half of the duopoly. Comparable quality but adding it would over-diversify; SNPS adequately captures the EDA theme. CDNS remains on watchlist.

8. AI Data Center Fabric — 4% (ANET)

Status: MODEST TRIM — 5% → 4%.

Arista Q1 2026 beat: revenue $2.709B (+35.1% YoY) vs $2.61B est. **However, full-year 2026 revenue guidance raised to $11.5B (+27.7%) — slightly below analyst high-end (28–30%).** Doubled the 2026 AI revenue target to $3.5B. Announced XPO (extended pluggable optics) for liquid-cooled AI datacenters. CEO Ullal flagged supply constraints across wafers, silicon, CPUs, optics, and memory as a "1-2 year phenomenon" — multi-year purchase commitments now $8.9B, inventory $2.38B; **margin pressure ahead from premium component procurement costs.**

The thesis is still valid (Ethernet displacing InfiniBand; Arista is the leader), but the slightly soft guidance, supply constraints, and the rise of Nvidia Spectrum-X as a credible competitor justify a modest trim from 5% → 4%. The freed 1% goes to the optical layer (COHR), which is upstream of switching and where pricing power is currently sharper.

Alternatives evaluated: - Cisco (CSCO): legacy enterprise focus; weak AI-cluster narrative. Worse. - Juniper / HPE Networking: weak AI-cluster narrative. Worse.

9. AI-Native Security — 3% (CRWD)

Status: UNCHANGED.

CrowdStrike Q4 FY26: ARR $5.25B (+24% YoY), **net new ARR $331M (+47% YoY record)**, full-year FY26 net new ARR >$1B for the first time. Q4 revenue $1.31B (+23%); GAAP net income $38.7M (vs -$86.3M Q4 FY25 loss); FCF $376.4M. **AI Detection and Response (AIDR) adoption +5x QoQ.** Management framing AI security as "mission-critical infrastructure across every layer from GPU to agent to prompt," and targeting $20B ARR by FY36.

The 3% allocation set in V8.3 (up from 2% in V7.3) was prescient. Net new ARR re-acceleration to +47% is a genuinely impressive datapoint in a slowing software market. Hold 3%.

Alternatives evaluated: - Palo Alto Networks (PANW): platform leader but more network-centric (less aligned with AI agent threats). Worse for this thesis. - Zscaler (ZS): cloud-native, but smaller AI-native security narrative. Worse.

10. AI-Native Operations — 2% (PLTR)

Status: UNCHANGED — but conviction has risen.

Palantir Q1 2026 was a blowout: **revenue $1.63B (+85% YoY) — fastest growth since IPO.** US commercial $595M; raised FY26 guide to $7.65–7.66B (+71% YoY) and US commercial guidance to **+120% YoY**. FY26 FCF guide $4.2–4.4B. Q2 guide $1.8B. US government revenue +84% YoY. Net income quadrupled to $870.5M.

Valuation remains the challenge — PLTR trades at >150x forward earnings — and the GPT5.5 input report explicitly recommends trimming to 1%. We disagree: a 2% position is sized appropriately for the volatility (it's already the smallest meaningful core position outside hedges), and AIP adoption is genuinely accelerating. **Hold 2%.** The "AI operating system" thesis is correct.

11. AI-Enabled Defense — 2% (AVAV 1%, KTOS 1%)

Status: UNCHANGED.

Kratos (KTOS): Q1 2026 revenue $371M (+22.6%; +15.8% organic); record backlog **$2B (up from $1.6B at year-end and $1.2B at Q4 2025)**; book/bill 1.6x; **$14B opportunity pipeline**. KUS (unmanned) revenue +30.9% organic on Valkyrie demand. The Northrop Grumman + Kratos team won the **U.S. Marine Corps MUX TACAIR CCA contract**, integrating Valkyrie with Northrop's mission kit. 2026 guide $1.7–1.76B (+15–19% organic).

AVAV: Fiscal Q3 2026 revenue $408M (+143% YoY), record funded backlog $1.1B, bookings $2.1B (book/bill 1.6x for first 9 months). **However, $151.3M goodwill impairment charge** from a stop-work order on the BADGER program (Space Force SCAR). Gross margin fell to 24% (from 38%). Forward outlook: management still expects record Q4 revenue.

KTOS is firing on all cylinders; AVAV has near-term execution risk from the BADGER impairment but the structural Switchblade / Replicator demand is intact. The 1%/1% split correctly hedges between the two — neither is large enough to be portfolio-defining, both capture meaningful upside if the "affordable mass" doctrine continues to receive procurement funding. Hold 2%.

Alternatives evaluated: - Lockheed Martin (LMT) / Northrop Grumman (NOC): high-quality but lower-beta; less concentrated exposure to autonomous systems. Worse for our specific thesis. Continuity Fund handles broad defense. - ITA ETF: see Cross-Sleeve Candidates.

12. Bio-Risk Hedge — 1% (MRNA)

Status: UNCHANGED.

GPT5.5 input recommends selling MRNA and replacing with IBB. We disagree because the **role of MRNA is asymmetric tail-risk insurance**, not broad biotech exposure. In the base case, MRNA treads water (and that's fine — 1% is small). In the tail case (engineered pathogen, AI-enabled

bioterrorism, novel pandemic), MRNA's mRNA platform provides the only viable rapid-response path and would experience explosive non-linear repricing.

IBB (broad biotech ETF) does not have the same asymmetric tail-payoff profile. It captures the median biotech outcome, which is exactly what we don't want from this slot. Hold 1%.

13. Strategic Materials — 3% (MP)

Status: UNCHANGED.

MP Materials Q1 2026 was a structural milestone: **NdPr production +63% YoY, sales +117% YoY; revenue $90.6M (+49%); Adjusted EBITDA $36.6M (vs -$2.7M).** $42.3M of PPA income (DoD 10-yr $110/kg NdPr price floor). Broke ground on the "10X" magnet facility (~10,000 metric tons annual capacity); Mountain Pass heavy rare earth separation commissioning Q2 2026.

The $400M DoD equity stake + 10-year price floor + 10-year magnet offtake provides a genuinely de-risked operational floor. The geopolitical hedge value is rising as China continues to weaponize rare earth exports. Hold 3%.

14. Domestic Foundry — 3% (INTC)

Status: MODEST TRIM — 4% → 3%.

Intel 18A is making real yield progress (CEO Tan reports **7-8% monthly yield gains**) and the 18A-P variant (8% higher performance per watt) is receiving "inbound interest" from foundry customers. Q1 2026 Intel Foundry revenue reached $5.4B. **However:** specific external customer volume commitments are still expected in **2H 2026 / 1H 2027**. Apple and Nvidia "inbound interest" reported in earlier cycles has not yet materialized into named volume contracts in the Q1 2026 filings.

The trim from 4% → 3% reflects this gap. INTC remains a credible Taiwan-contingency hedge and a sovereign-foundry option, but the optionality has not yet converted to confirmed customer wins. If a major design win is announced (Nvidia, Apple, or an AI-cloud customer at scale), consider increasing back to 4-5%.

Alternative evaluated: Convert INTC to LEAPS call options as Aschenbrenner does. This is a legitimately convex structure — but LEAPS strategies are explicitly held off until the user is comfortable with options mechanics. Keep as equity for now.

15. Electrical Equipment — 4% (GEV)

Status: INCREASE — 3% → 4%.

GE Vernova Q1 2026 was the strongest quarterly print of any portfolio holding by raw scale: **total backlog $163B (+$13B sequentially), orders $18.3B (+71% organic), book/bill ~2x.** Gas turbine backlog grew from 83 GW to **100 GW**, with 21 GW of new gas equipment contracts in a single quarter (US, Vietnam, Mexico, Brazil, Canada). Equipment backlog $76B (+67% YoY). Electrification orders $7.1B (+86% organic); equipment backlog there $38.6B (+$16.6B YoY) including $5B from the Prolec GE acquisition. Targeting 110 GW gas turbine backlog by year-end 2026 and 24 GW/year manufacturing capacity by 2028.

GE Vernova is the apex monopoly in gas turbines and high-voltage power transformers, the two pieces of equipment that limit how fast new datacenter capacity can come online. 128-week transformer lead times mean the order book is essentially physically locked in. GPT5.5's input report flagged GEV as "valuation-extended" and recommended trimming to 2%; Gemini's input report recommended increasing to 5%. We side closer to Gemini: when backlog is growing at a structural deficit, valuation premium is justified. Increase to 4%.

16. Nuclear Fuel — 2% (CCJ)

Status: UNCHANGED.

Cameco Q1 2026: net earnings $131M; adjusted EBITDA $509M (uranium $423M + Westinghouse share $122M, both up YoY); maintained 2026 production guidance 19.5–21.5M lbs U3O8. Westinghouse contributed positively (Cameco's share of net loss narrowed to -$46M from -$62M, with adjusted EBITDA contribution rising to $122M). Received $49M distribution from Westinghouse during the quarter.

The uranium structural deficit remains intact (190M lb demand vs ~135M lb mine supply globally). Cameco's deliberate supply discipline (producing 19.5–21.5M lbs at McArthur River vs 25M lb licensed capacity) preserves long-term pricing power. Hold 2%.

17. Grid-Bypass Generation — 3% (BE) — NEW SLEEVE

Status: NEW POSITION — 0% → 3%.

This is the single most important addition in the V9.0 cycle.

Bloom Energy (BE) manufactures solid-oxide fuel cells (SOFCs) that generate electricity on-site using natural gas, biogas, or hydrogen blends. The technology is uniquely positioned for AI datacenter deployment because: - It bypasses the 5–7 year regulated-grid interconnection queue. - It scales modularly to gigawatt-class campuses. - It natively produces **DC power**, which aligns with Nvidia's 800V DC rack architecture trend, eliminating ~10–15% of energy losses from AC/DC conversion. - It can run on natural gas, biogas, or hydrogen — full fuel flexibility.

Q1 2026 results were the inflection point: **revenue $751M (+130% YoY)**; net income +$70.65M (vs prior loss); raised full-year 2026 revenue guidance to $3.40–3.80B (>50% growth); total backlog ~$20B.

The defining event was the **Oracle Project Jupiter agreement (April 27, 2026)**: up to **2.8 GW of Bloom fuel cells** to power Oracle's AI datacenter campus in Doña Ana County, NM. Initial 1.2 GW already contracted with deployment underway. This deal **replaces previously planned gas turbines and diesel generators** with continuous fuel-cell power — proving that fuel cells are now considered primary baseload, not backup.

The 3% allocation is meaningful but not aggressive (Aschenbrenner's fund has 16–22% in BE — we are intentionally smaller given concentration discipline). It captures genuine grid-bypass exposure that no other portfolio holding provides.

Funding: From the trims to TSM, VRT, DELL, ANET, INTC, and CEG.

Alternatives evaluated: - FuelCell Energy (FCEL), Plug Power (PLUG): inferior unit economics, weaker hyperscaler relationships, higher dilution risk. Worse. - ETN (Eaton): captures some of the same

datacenter electrical-equipment thesis, but is much more diversified and not a fuel cell pure-play. Comparable for general AI-power exposure but Worse for the specific grid-bypass thesis. (See Cross-Sleeve section.)

18. Optical Fabric — 2% (COHR) — NEW SLEEVE

Status: NEW POSITION — 0% → 2%.

Coherent (COHR) manufactures optical components and transceivers — Indium Phosphide lasers (EML, narrow-linewidth), 800G and 1.6T transceivers, and (importantly) co-packaged optics (CPO). Q3 FY26 results: **revenue $1.81B (+21% YoY)**; non-GAAP gross margin 39.6% (+105 bps YoY); non-GAAP EPS $1.41 (+55% YoY). Data Center & Communications segment = 75% of revenue and grew **>40% YoY**. Within that, Communications grew +60% YoY on data-center interconnect (DCI).

The defining event: **Nvidia took a $2B equity investment in Coherent + signed a multi-year supply agreement.** This is the strongest possible anchor-customer validation for the optical fabric thesis. Initial scale-out CPO revenue ramps in H2 calendar 2026; scale-up CPO in H2 2027. **OCS market opportunity raised to >$4B; CPO TAM seen as >$15B.**

Coherent is doubling InP capacity by Q4 FY26 (one quarter ahead of plan). The narrow choice between Coherent and Lumentum (LITE):

Criterion	LITE	COHR
Pure InP/optical play	Yes	Mostly (also industrial lasers)
Q3 FY26 revenue	$808M	**$1.81B**
Q3 FY26 YoY revenue growth	+90%	+21%
Q3 FY26 op margin	32.2%	33% (non-GAAP)
Anchor customer	Hyperscalers	**Nvidia ($2B equity stake)**
Diversification	Pure-play, more concentrated	Broader (industrial lasers, communications)
Risk profile	Higher concentration	Lower concentration

We choose **Coherent over Lumentum** because the Nvidia equity stake provides structural validation, and the broader product mix reduces single-cycle risk. Lumentum's faster YoY growth is impressive but reflects a smaller revenue base.

Funding: From the trims to TSM, VRT, DELL, ANET, INTC, and CEG.

19. Moonshot — 0%

Status: UNCHANGED — held at 0% pending LEAPS understanding.

The compelling LEAPS candidates from the May 5 Gemini input — **CoreWeave (CRWV), Core Scientific (CORZ), Iris Energy (IREN)** — are all real and validated: - CRWV: $99.4B backlog, $21B Meta deal, 91% Q1 revenue growth. - CORZ: 243 MW delivered to CoreWeave at 80–85% gross margins; 4.5 GW pipeline; $3.3B project bond closed. - IREN: $9.7B Microsoft AI cloud deal, $5.8B Dell GPU supply agreement, 4.5 GW secured grid power, +613% YoY.

These are the **prime LEAPS targets** when the Moonshot sleeve is reactivated. They share a common profile (energized real estate + GPU hosting) and are highly volatile equity stories — exactly the kind of asymmetric exposure that LEAPS structures are designed for. **For now, hold the Moonshot at 0%** until LEAPS strategies are understood; revisit after the user has educated themselves on debit call spreads, expiration management, and roll discipline.

Part V: Cross-Sleeve Candidate Evaluation

Per the prompt template, we evaluate **all credible alternatives** surfaced by GPT5.5 / Gemini inputs and current market developments — including those that don't fit existing sleeves.

Ticker	Candidate Theme	Why Considered	Better/Comparable/ Worse	Add?	Why / Why not
BE	Grid-bypass fuel cells	Oracle 2.8 GW deal; +130% YoY rev; Aschenbrenner top position	**Better** (no equivalent in V8.3)	**Yes**	New sleeve at 3%
COHR	Optical fabric / CPO	Nvidia $2B equity stake; 75% rev from DC&C	**Better** than ANET-only for optical	**Yes**	New sleeve at 2%
LITE	Optical fabric	+90% YoY rev; pure InP play	Comparable to COHR but more concentrated	No	COHR chosen instead
CRWV	GPU cloud (CoreWeave)	$99.4B backlog; +91% rev; $21B Meta deal	Comparable but option-suitable, not equity	Defer to Moonshot	LEAPS target
CORZ	Compute landlord (Texas/ OK)	243 MW @ 80-85% margins; 4.5 GW pipeline	Comparable but option-suitable	Defer to Moonshot	LEAPS target
IREN	Compute landlord + AI cloud	MS $9.7B deal; +613% YoY; 4.5 GW grid power	Comparable but option-suitable	Defer to Moonshot	LEAPS target
TLN	Pure nuclear (AWS PPA)	$1.13B Q1 revenue (+190%); $63M net income	Worse vs CEG (single-asset risk; 2.2 GW vs CEG 22 GW)	No	CEG/VST already cover nuclear
NRG	Integrated power + retail	Q1 results decent; LS Power acquisition	Worse (consumer retail dilutes thesis)	No	Out of scope
NEE	Renewables + utility	World's largest wind/ solar	Worse (renewables are supplement, not baseload)	No	Captured via Continuity Fund
ETN	Diversified electrical equipment	Data center orders +240% YoY; $11B M&A	Comparable to GEV+PWR but more diluted	No	GEV+PWR more concentrated; ETN's aerospace mix dilutes thesis
AMD	GPU competition	Real but no unique bottleneck	Worse	No	Captured indirectly via TSM/MU

Ticker	Candidate Theme	Why Considered	Better/Comparable/ Worse	Add?	Why / Why not
MRVL	Custom silicon adjunct	Smaller scale than AVGO	Worse	No	AVGO is the cleaner play
CDNS	EDA duopoly	Comparable to SNPS	Comparable	No	SNPS adequately captures the theme
LRCX / KLAC	WFE oligopoly	Quality oligopolists	Comparable	No	ASML+AMAT adequately captures WFE
HPE	AI server competitor	Direct DELL competitor	Worse (same business model issues)	No	Pass
SMCI	AI server	Governance issues	Much worse	No	Explicitly excluded
ORCL	"Challenger hyperscaler"	Strong cloud growth; Bloom Energy partner	Worse (doesn't own bottleneck; is a buyer)	No	Captured via Continuity Fund
PANW	Network security	Less aligned with AI-agent threats	Worse than CRWD	No	CRWD better fits AI thesis
ZS	Cloud security	Smaller AI-native narrative	Worse than CRWD	No	Pass
LMT / NOC	Defense primes	Lower-beta, less autonomous-systems exposure	Worse for affordable-mass thesis	No	Continuity Fund handles broad defense
SOXX	Broad semis ETF	GPT5.5 recommended adding 6%	Worse (dilutes thesis; redundant with TSM/ ASML/NVDA/MU)	No	Continuity Fund handles broad semis
XLU	Broad utilities ETF	GPT5.5 recommended adding 4%	Worse (dilutes thesis)	No	CEG/VST/PWR are the high-conviction names
ITA	Aerospace/ defense ETF	GPT5.5 recommended adding 2%	Worse (dilutes affordable-mass thesis)	No	Continuity Fund handles broad defense
IBB	Biotech ETF	GPT5.5 recommended replacing MRNA	Worse for tail-risk asymmetric payoff	No	MRNA is insurance, not broad biotech exposure
IBM	Quantum + AI	Some interesting moves but slow exec	Worse	No	Pass
SNOW	Data cloud	Compute-buyer thesis	Worse	No	Captured via Continuity Fund

Part VI: Recommended Optimized Allocation

Final Event Horizon Fund Allocation (V9.1 — May 10, 2026, ACTIVE — supersedes V9.0)

Sleeve	Ticker	V8.3 (Mar 31)	V9.0 (May 10)	V9.1 (Active)	V9.0→V9.1	Rationale
Energy / Power	CEG	12%	11%	**9%**	**-2%**	Both followups: single-name/event risk too high at 11%; FERC co-location friction
	VST	3%	4%	**6%**	**+2%**	Both followups: faster growth + Fitch IG upgrade + oil-shock resilience; funded by CEG
	PWR	3%	3%	**4%**	**+1%**	GPT strongly recommends; record $48.5B backlog; cleanest grid-buildout toll road; funded by DELL
Foundry	TSM	12%	11%	**11%**	—	V9.0 11% balanced (Gemini wanted 12%, GPT wanted 9.5% — hold midpoint)
WFE	ASML	8%	8%	**8%**	—	EUV monopoly intact; raised guide €36–40B
	AMAT	4%	4%	**4%**	—	Diversified WFE + U.S. geopolitical hedge
Rack-Scale & Cooling	VRT	8%	6%	**6%**	—	Already trimmed in V9.0; conflict between followups (Gemini +2%, GPT -1.5%) — hold
	DELL	5%	4%	**2%**	**-2%**	Both followups: integrator economics, weakest bottleneck purity
Accelerators	NVDA	6%	6%	**6%**	—	$78B Q1 FY27 outlook; Rubin shipping H2 FY27 with 10x cost reduction
	AVGO	3%	3%	**3%**	—	Q1 AI rev $8.4B (+106%); >$100B AI rev visibility for 2027
Memory	MU	6%	6%	**6%**	—	HBM thesis intact; conflict between followups (Gemini +1, GPT -1) — hold midpoint
EDA	SNPS	5%	5%	**5%**	—	Core EDA toll road; DA segment +96% YoY; rejected Gemini's unjustified trim
Networking	ANET	5%	4%	**4%**	—	Already trimmed in V9.0; further trim not warranted given Spectrum-X context
Security	CRWD	3%	3%	**3%**	—	Net new ARR +47% YoY; AIDR +5x QoQ; targeting $20B ARR by FY36
Operations	PLTR	2%	2%	**2%**	—	Genuine ops/control rail; Q1 +85% growth; conflict not decisive
Defense	AVAV	1%	1%	**1%**	—	Hold despite BADGER goodwill impairment; structural demand intact
	KTOS	1%	1%	**1%**	—	Record $2B backlog; CCA contract awarded with Northrop Grumman
Bio-Risk	MRNA	1%	1%	**1%**	—	Tail-risk insurance; asymmetric payoff hedge
Materials	MP	3%	3%	**3%**	—	Q1 +49% rev; DoD price floor + 10X magnet facility groundbreaking

Sleeve	Ticker	V8.3 (Mar 31)	V9.0 (May 10)	V9.1 (Active)	V9.0→V9.1	Rationale
Domestic Foundry	INTC	4%	3%	**3%**	—	Hold V9.0 weight; rejected Gemini's LEAPS conversion (no LEAPS per user rule)
Electrical Equipment	GEV	3%	4%	**5%**	**+1%**	Both followups: $163B backlog + 128-week transformer lead times; funded by DELL
Nuclear Fuel	CCJ	2%	2%	**2%**	—	Q1 EBITDA $509M; Westinghouse contributed positively
Grid-Bypass	BE	0%	3%	**3%**	—	Hold; rejected Gemini chase to 6%, rejected GPT trim to 2% — current sizing balanced
Optical Fabric	COHR	0%	2%	**2%**	—	Hold; rejected Gemini swap to LITE — COHR has stronger Nvidia $2B validation
Moonshot	—	0%	0%	**0%**	—	Held at 0% pending LEAPS understanding; CRWV/CORZ/IREN as top targets
Total		**100%**	**100%**	**100%**	**0%**	**V9.1 Checksum verified: 9+6+4+11+8+4+6+2+6+3+6+5+4+3+2+1+1+1+3+3+5+2+3+2 = 100.0%**

Summary of Follow-up Changes from V9.0 → V9.1 (Active)

Change	From (V9.0)	To (V9.1)	Source of Funds	Rationale
CEG trim	11%	**9%**	(frees 2.0%)	Both followups: 11% single-name/event risk too high; FERC co-location friction
DELL trim	4%	**2%**	(frees 2.0%)	Both followups: integrator economics, not a bottleneck owner
VST increase	4%	**6%**	CEG trim (-2%)	Both followups: faster growth + Fitch IG upgrade + oil-shock resilience
GEV increase	4%	**5%**	DELL trim (-1%)	Both followups: $163B backlog + 128-week transformer lead times
PWR increase	3%	**4%**	DELL trim (-1%)	GPT strongly recommends; record $48.5B backlog; cleanest grid-buildout toll road

V9.1 Net effect: Power chain (CEG+VST+PWR+GEV+BE+CCJ) sharpens from 27% → **29%** (highest-conviction bottleneck reinforced); Rack-Scale (VRT+DELL) trims from 10% → **8%** (DELL right-sized to integrator-appropriate weight). All other 18 holdings unchanged. **V9.1 sleeve totals:** Power Generation **19%** (up 1%), Semis Foundry/WFE 23%, Rack-Scale **8%** (down 2%), Accelerators 9%, Memory 6%, EDA 5%, Networking 4%, Security 3%, Operations 2%, Defense 2%, Bio-Risk 1%, Materials 3%, Domestic Foundry 3%, Electrical Equipment **5%** (up 1%), Nuclear Fuel 2%, Grid-Bypass 3%, Optical Fabric 2%, Moonshot 0%.

Summary of All Changes from V8.3 → V9.0 (Historical — superseded by V9.1)

Change	From	To	Source of Funds	Rationale
BE (NEW)	0%	3%	Various trims	Oracle 2.8 GW; grid-bypass thesis validated; +130% YoY rev

Change	From	To	Source of Funds	Rationale
COHR (NEW)	0%	2%	Various trims	Nvidia $2B equity stake; optical bottleneck migrated upstream
VST increase	3%	4%	CEG trim	Q1 huge beat; Investment-Grade upgrade
GEV increase	3%	4%	TSM/VRT/DELL trim	Backlog $163B; gas turbine backlog 100 GW
CEG trim	12%	11%	Funds VST	Modest concentration management; soft 2026 EPS guide
TSM trim	12%	11%	Funds new sleeves	Modest concentration trim; thesis still intact
VRT trim	8%	6%	Funds new sleeves	Valuation extended (~88x trailing)
DELL trim	5%	4%	Funds new sleeves	Systems integrator, not bottleneck owner
ANET trim	5%	4%	Funds COHR	Supply constraints + 2026 guide slightly soft
INTC trim	4%	3%	Funds new sleeves	No named customer volume commits yet

V9.0 net effect (historical): The portfolio adds two genuinely new bottleneck exposures (grid-bypass via BE; optical fabric via COHR) while modestly trimming valuation-extended winners and concentration-heavy positions. All other 18 holdings remain unchanged. V9.0 sleeve totals: Power Generation 18%, Semis Foundry/WFE 23%, Rack-Scale 10%, Accelerators 9%, Memory 6%, EDA 5%, Networking 4%, Security 3%, Operations 2%, Defense 2%, Bio-Risk 1%, Materials 3%, Domestic Foundry 3%, Electrical Equipment 4%, Nuclear Fuel 2%, Grid-Bypass 3% (NEW), Optical Fabric 2% (NEW), Moonshot 0%.

Part VII: Per-Ticker Decision Framework (V9.1 Active)

Per the prompt template requirement, here is the explicit per-ticker decision matrix. **The V9.1 % column is the active allocation.** V9.0 column is retained for audit/historical comparison only.

Ticker	V8.3 %	V9.0 %	V9.1 % (Active)	V9.0→V9.1 Action	Confidence (1-5)	Why	What would falsify this
CEG	12%	11%	**9%**	**Reduce -2%**	5	Both followups: 11% single-name/event risk too high; FERC co-location friction; concentration management	Major new ≥1 GW PPA at premium pricing; or DOE policy shift
VST	3%	4%	**6%**	**Increase +2%**	5	Both followups: faster growth + Fitch IG upgrade + oil-shock resilience; gas+nuclear hedge	Wholesale price collapse; or nuclear plant operating issue
PWR	3%	3%	**4%**	**Increase +1%**	5	GPT-strong: record $48.5B backlog; cleanest grid-buildout toll road; doubling transformer capacity	Material slowdown in grid spending

Ticker	V8.3 %	V9.0 %	V9.1 % (Active)	V9.0→V9.1 Action	Confidence (1-5)	Why	What would falsify this
TSM	12%	11%	**11%**	Keep	4	V9.0 11% balanced (Gemini wanted 12%, GPT wanted 9.5% — hold midpoint); thesis strengthened	Geopolitical event in Taiwan; or Intel external customer wins
ASML	8%	8%	**8%**	Keep	5	Raised guide to €36–40B; EUV monopoly intact	Credible non-EUV 2nm patterning alternative emerges (very low probability)
AMAT	4%	4%	**4%**	Keep	4	U.S. WFE hedge; CHIPS Act beneficiary	China export control escalation that hits AMAT specifically
VRT	8%	6%	**6%**	Keep	3	Already trimmed in V9.0; conflict between followups (Gemini +2%, GPT -1.5%) — hold	Schneider Electric or Eaton wins majority of new hyperscaler cooling RFPs
DELL	5%	4%	**2%**	**Reduce -2%**	4	Both followups: integrator economics, weakest bottleneck purity; HPE competition	AI server gross margin expansion above 13%
NVDA	6%	6%	**6%**	Keep	5	Core compute; conflict between followups (GPT +1, Gemini hold) — hold; $78B Q1 FY27 guide	Hyperscalers shift training (not just inference) to custom ASICs for two consecutive generations
AVGO	3%	3%	**3%**	Keep	5	$100B+ 2027 AI revenue visibility from chips alone	Loss of Google or Meta custom-ASIC partnership
MU	6%	6%	**6%**	Keep	5	HBM thesis intact; conflict between followups (Gemini +1, GPT -1) — hold midpoint	Yield issue on HBM4 vs SK Hynix
SNPS	5%	5%	**5%**	Keep	4	Core EDA toll road; DA segment +96% YoY; rejected Gemini's unjustified trim	CDNS materially gains share for two consecutive years
ANET	5%	4%	**4%**	Keep	3	Already trimmed in V9.0; further trim not warranted given Spectrum-X context	Sustained share gains in 800G/1.6T despite supply issues
CRWD	3%	3%	**3%**	Keep	5	Net new ARR +47% YoY; AI security non-discretionary	ARR growth deceleration below 20% for two consecutive quarters
PLTR	2%	2%	**2%**	Keep	4	Genuine ops/control rail; Q1 +85% growth; conflict not decisive	Sustained <40% commercial growth deceleration

Ticker	V8.3 %	V9.0 %	V9.1 % (Active)	V9.0→V9.1 Action	Confidence (1-5)	Why	What would falsify this
AVAV	1%	1%	**1%**	Keep	3	Switchblade/Replicator backlog intact despite BADGER impairment	Loss of major P550 program-of-record
KTOS	1%	1%	**1%**	Keep	4	Record $2B backlog; CCA contract real	Loss of CCA tranche awards to competitors
MRNA	1%	1%	**1%**	Keep	4	Tail-risk asymmetric insurance; both followups agree small	A direct competitor (BNTX) demonstrates faster speed-to-vaccine
MP	3%	3%	**3%**	Keep	4	$42M PPA income; 10X magnet facility groundbreaking	Major rare earth substitution breakthrough
INTC	4%	3%	**3%**	Keep	3	Hold V9.0 weight; rejected Gemini's LEAPS conversion (no LEAPS per user rule)	Apple or Nvidia announces specific volume foundry contract
GEV	3%	4%	**5%**	**Increase +1%**	5	Both followups: $163B backlog (+$13B QoQ); 100 GW gas turbine backlog; transformer monopoly	Hyperscaler capex slowdown that extends to 2027 turbine orders
CCJ	2%	2%	**2%**	Keep	4	Q1 EBITDA $509M; Westinghouse improving	Major reactor design substitution that makes Westinghouse less relevant
BE	0%	3%	**3%**	Keep	4	Hold; rejected Gemini chase to 6%, rejected GPT trim to 2% — current sizing balanced after vertical move	Oracle delays/cancels Project Jupiter; or hydrogen alternative emerges with better unit economics
COHR	0%	2%	**2%**	Keep	4	Hold; rejected Gemini swap to LITE — COHR has stronger Nvidia $2B validation	Silicon photonics breakthrough that dramatically reduces transceiver demand
Moonshot	0%	0%	**0%**	Keep	5	LEAPS strategies still pending user education; rejected CORZ/IREN adds (deferred to LEAPS sleeve)	User completes LEAPS education; reactivate at 8–10% with CRWV/ CORZ/IREN structures

Part VIII: Stocks NOT in the Portfolio — Why They Stay Out

Hyperscalers (GOOGL, MSFT, AMZN, META)

Buyers of scarce resources, not providers. Already held via the Continuity Fund (FSKAX / FTIHX). Adding them dilutes the bottleneck thesis.

AMD (AMD)

Credible #2 GPU but doesn't own a unique structural bottleneck. Its TSMC capacity is captured indirectly via TSM; its HBM is captured via MU.

Lumentum (LITE)

Strong optical play, but Coherent (COHR) provides the same thematic exposure with stronger anchor-customer validation (Nvidia $2B equity) and a more diversified product mix.

CoreWeave (CRWV), Core Scientific (CORZ), Iris Energy (IREN)

The compute-landlord thesis is real and powerful. These names are deferred to the Moonshot/LEAPS sleeve when reactivated — their high equity volatility makes them better suited to convex option structures than to direct equity allocation.

Talen Energy (TLN)

Pure-play nuclear with the AWS Susquehanna PPA, but a single-asset concentration risk (2.2 GW). CEG (22 GW) and VST (6.4 GW nuclear) provide more diversified nuclear exposure.

Eaton (ETN)

Data center orders +240% YoY and $11B in Boyd Thermal + Ultra PCS acquisitions are impressive, but ETN's aerospace and broader industrial mix dilutes the AI-power thesis. GEV + PWR + VRT capture the targeted exposure more cleanly.

Cadence (CDNS)

Strong EDA duopolist alongside SNPS. SNPS at 5% adequately captures the EDA theme; CDNS would over-diversify without adding a new bottleneck.

Lam Research (LRCX), KLA (KLAC)

Strong WFE oligopolists, but ASML (EUV monopoly) + AMAT (broadest portfolio + U.S. footprint) adequately capture the WFE thesis.

Oracle (ORCL)

Challenger hyperscaler, but it's a *buyer* of compute and power, not a provider. Already captured via Continuity Fund.

Super Micro (SMCI)

Explicit exclusion: governance concerns, margin compression, and the market shift toward integrators with global scale (DELL, HPE) rather than speed-to-market specialists.

Broad ETFs (SOXX, XLU, ITA, IBB)

GPT5.5 input recommends adding these as ballast. We disagree: the Continuity Fund (FSKAX 80% / FTIHX 20%, ~50% of total investments) already provides broad market exposure including semis, utilities, defense, and biotech. Adding broad ETFs to the EHF would dilute the concentrated bottleneck thesis without improving risk-adjusted returns.

Part IX: Risk and Invalidation Dashboard

Tier 1 Risks (Portfolio-Level)

1. Taiwan Contingency - **Risk:** PRC military action against Taiwan disrupting TSMC output - **Impact:** Severe (TSM 11%, indirect effects across NVDA/AVGO/MU) - **Hedge:** INTC (3%) becomes most valuable company on earth; ASML, AMAT, MP, defense names benefit from accelerated onshoring - **Action trigger:** Official U.S. intelligence warning of "imminent" action → reduce TSM to 5%, rotate into INTC (+3%), defense (+2%), MP (+1%)

2. Hyperscaler Capex Plateau or Reversal - **Risk:** AI revenue model fails to materialize, hyperscalers cut capex aggressively in 2027 - **Impact:** High across power, foundry, memory, networking, optical - **Assessment:** Currently low-probability. $725-830B 2026 capex was REVISED UP from $660-690B in March; hyperscaler earnings (MSFT, GOOGL, AMZN, META) all reaffirmed or raised 2026 capex. Phantom backlog risk exists but is partially mitigated by 20-25% upfront slot reservation deposits. - **Monitor:** Quarterly hyperscaler capex guidance. If 2 consecutive quarters show meaningful deceleration, reassess.

3. PJM Capacity Auction / FERC Wholesale Cap Risk - **Risk:** FERC or Congress imposes wholesale electricity caps or renegotiates existing nuclear PPAs, undermining premium nuclear pricing - **Impact:** Moderate to severe for CEG (11%), VST (4%) - **Assessment:** April 16, 2026 FERC ruling was actually slightly positive (clarified co-location pathways). Bilateral PPAs (Microsoft, Meta) are private contracts not subject to wholesale caps. Risk is real but lower than feared at March 2026. - **Trim trigger:** If FERC broadly prohibits direct nuclear PPAs to tech companies → reduce CEG to 7%, VST to 3%; rotate into BE (grid-bypass; benefits from regulated-grid friction).

4. Macro / Interest Rate / Recession Shock - **Risk:** Severe recession or credit crisis reduces all tech spending - **Impact:** Beta-adjusted, the EHF would suffer ~1.5-2x market decline in a sharp drawdown - **Hedge:** CEG/PWR/VST provide some utility-like defensive characteristics. Continuity Fund (50% of total investments) in broad indexes provides cushion.

Tier 2 Risks (Sleeve-Level)

5. Nvidia Custom ASIC Displacement - **Risk:** Hyperscalers shift training (not just inference) to custom ASICs - **Impact:** Reduces NVDA growth; benefits AVGO - **Assessment:** Not happening in 2026-2027 window. Training remains NVDA monopoly. Custom ASICs for inference actually expand total market.

6. Optical Substitution Risk (Silicon Photonics) - **Risk:** Co-packaged optics or silicon photonics breakthroughs reduce demand for InP transceivers - **Impact:** Hurts COHR (and broader optical thesis) - **Assessment:** Coherent and Lumentum are themselves leading the CPO transition; the substitution risk is more about market share than market size.

7. Bloom Energy Customer Concentration / Technology Risk - **Risk:** Oracle Project Jupiter delayed; technology fails to scale to gigawatt-class reliability - **Impact:** Severe for BE (3%) — could cut position by half overnight - **Assessment:** Customer concentration is real (Oracle is anchor); but $20B backlog provides multi-customer visibility. Hydrogen-blend optionality preserves long-term economics. - **Trim trigger:** Oracle announces material delay or alternative power vendor → reduce BE to 1.5%.

Top 5 Invalidation Triggers (Portfolio-Wide)

1. **Hyperscaler capex deceleration** for 2 consecutive quarters
2. **Taiwan contingency** intelligence warning
3. **FERC wholesale price cap** or nuclear PPA prohibition
4. **METR task-horizon trend break** (no improvement for 6+ months)
5. **Broad recession** with -20%+ market drawdown

Part X: Performance Expectations Under V9.1 (Active)

Projected Annual Returns by Sleeve (2026E - 2028E)

Sleeve	V9.1 Weight	2026E	2027E	2028E
Energy / Power Gen (CEG 9% + VST 6% + PWR 4%)	**19%**	+25–40%	+30–45%	+20–30%
Foundry (TSM)	11%	+25–35%	+25–35%	+15–25%
WFE (ASML/AMAT)	12%	+25–35%	+25–35%	+18–25%
Rack-Scale (VRT 6% + DELL 2%)	**8%**	+30–45%	+25–35%	+18–25%
Accelerators (NVDA/AVGO)	9%	+30–45%	+28–40%	+18–28%
Memory (MU)	6%	+35–50%	+25–40%	+15–25%
EDA (SNPS)	5%	+20–28%	+22–28%	+18–22%
Networking (ANET)	4%	+20–30%	+25–35%	+15–25%
Security (CRWD)	3%	+25–35%	+30–40%	+25–30%
Operations (PLTR)	2%	+20–35%	+25–40%	+20–30%
Defense (AVAV/KTOS)	2%	+25–40%	+30–45%	+20–30%
Bio-Risk (MRNA)	1%	-10 to +20%	-5 to +25%	-5 to +30%

Sleeve	V9.1 Weight	2026E	2027E	2028E
Materials (MP)	3%	+20–35%	+20–30%	+15–25%
Domestic Foundry (INTC)	3%	+30–55%	+25–45%	+18–28%
Electrical Equip (GEV)	**5%**	+25–40%	+35–45%	+25–35%
Nuclear Fuel (CCJ)	2%	+15–25%	+25–35%	+25–35%
Grid-Bypass (BE)	3%	+30–60%	+35–55%	+25–40%
Optical Fabric (COHR)	2%	+25–45%	+30–45%	+20–30%
Weighted Portfolio	**100%**	**~28–40%**	**~28–40%**	**~20–30%**

> **V9.1 vs V9.0 weight changes:** Power Gen 18%→**19%** (+1), Rack-Scale 10%→**8%** (-2), Electrical Equip 4%→**5%** (+1). Net effect slightly tilts portfolio toward power chain, which has the strongest Q1 2026 earnings momentum and the most binding bottleneck.

Cumulative 3-year return (base case): ~110–145% (roughly 2.0–2.5x) **Cumulative 3-year return (bull case):** ~165–220% (roughly 2.6–3.2x) **With reactivated Moonshot LEAPS hitting:** ~210–290% (roughly 3.1–3.9x)

These projections assume the AI-2027 scenario broadly continues, hyperscaler capex remains elevated, and physical bottlenecks remain binding. **Note:** Forward returns are slightly more conservative than V8.3's projections because (a) several names are now at higher valuation starting points, and (b) the macro backdrop has flipped from oversold (March) to record highs (May), reducing near-term beta tailwind.

Part XI: Three-Fund Architecture Assessment

Is the 50/50 Split (Continuity / Event Horizon) Still Optimal?

The three-fund architecture remains sound and unchanged from V8.3:

1. **Foundation Liquidity** (SPAXX): Properly sized emergency fund + CMA spending. No changes needed.
2. **Continuity Fund (~50%)**: FSKAX 80% / FTIHX 20% in tax-advantaged accounts (Roth IRA, HSA, 401k with 5% match). This provides: - Broad market diversification the EHF deliberately lacks - Tax-advantaged compounding - Psychological stability during EHF volatility - Automatic exposure to hyperscalers, broad biotech, broad utilities, broad defense - The 5% 401k match capture is correct — never leave free money on the table
3. **Event Horizon Fund (~50%)**: The concentrated AI infrastructure bet, fully deployed across 24 names per V9.1 (active allocation).

Assessment: The 50/50 split is appropriate for a high-conviction multi-year time horizon. The Continuity Fund provides the floor; the EHF provides the ceiling.

One consideration unchanged from V8.3: As the EHF appreciates (which it has materially since September 2025 — power names alone have moved 50–100%+), the actual split likely sits above 50% EHF. Periodic rebalancing back toward 50/50 with new contributions to the Continuity Fund is prudent risk management. During the strongest phase of the AI buildout (2026–2028), letting the EHF run above 50% is acceptable if the thesis continues to validate.

Part XII: Conclusion

The Event Horizon Fund continues to be a well-constructed, thesis-coherent thematic portfolio. Q1 2026 results across the holdings — TSMC, ASML, GE Vernova, Vertiv, Quanta, Cameco, CrowdStrike, Palantir, Vistra, Dell, Coherent — were almost uniformly **better than expected**, with 2026 guidance being raised across the great majority of names. Hyperscaler capex was revised UP from $660-690B to $725-830B for 2026.

The thesis is more valid, not less.

What's changed is that the *location* of the binding bottleneck has migrated. The pure-compute and packaging bottlenecks (TSM CoWoS, HBM) are still tight but expanding aggressively. The newly binding bottlenecks are **front-of-meter and behind-the-meter power delivery** and **optical fabric inside hyperscale clusters**. The V9.0 cycle adds two new core positions to address those gaps: - **Bloom Energy (BE) at 3%** — grid-bypass via solid-oxide fuel cells; Oracle 2.8 GW Project Jupiter deal is the validating event. - **Coherent (COHR) at 2%** — optical fabric / co-packaged optics; Nvidia $2B equity stake is the anchor-customer signal.

These additions are funded by modest valuation/concentration trims to TSM, VRT, DELL, ANET, INTC, and CEG (1% each), redirected together with a +1% increase to VST and +1% to GEV. All other 18 holdings remain unchanged.

The portfolio remains positioned for what is shaping up to be the largest industrial buildout in human history. $725–830B in hyperscaler capex this year — exceeding the entire global semiconductor industry's 2025 revenue (~$600B) by 20–40% — flows directly through these companies. They are the foundries, equipment makers, power generators, fuel cells, transformers, gas turbines, memory suppliers, optical providers, and cooling engineers that make AI physically possible.

If the AI-2027 scenario plays out within even 50% of its projected trajectory, this portfolio should significantly outperform broad market indices over the 2026-2030 period.

Final V9.1 recommendation (Active): Deploy the V9.1 allocation gradually per the companion Deployment Strategy 2026-05-10 document (note: deployment file may need refresh to reflect V9.1 targets — see Part XIII for V9.0→V9.1 deltas). Hold. Monitor against the "must-be-true" thresholds documented per ticker. Rebalance only on genuine thesis invalidation, not on price volatility.

> **Reminder:** V9.1 supersedes V9.0 effective May 10, 2026 after follow-up reconciliation by Gemini + GPT5.5. See **Part XIII: V9.0 → V9.1 Reconciliation (Follow-up)** for full change rationale. V9.0 references in this document are retained only as audit/prior-cycle context and are no longer the active allocation.

Analysis prepared on May 10, 2026 by synthesizing the prior V8.3 cycle (March 31, 2026), GPT5.5 Pro Research (May 5, 2026), Gemini Pro Research (May 5, 2026), follow-up reconciliation by Gemini + GPT5.5 (May 10, 2026), and fresh stock-by-stock research across all 24 current holdings + key alternatives. All earnings figures sourced from company Q1 2026 / fiscal-equivalent reports as listed in the source links of the input research. Market data timestamped May 5–10, 2026. Where forward expectations are stated, they reflect the analyst's best assessment of company guidance, consensus estimates, and operating trajectory; they are not commitments and should be re-validated quarterly. Not financial advice.

Part XIII: V9.0 → V9.1 Reconciliation (Follow-up)

This section documents the follow-up reconciliation conducted on May 10, 2026 after independent review by Gemini Pro Research and GPT5.5 Pro Research. The full reconciliation analysis is in `Analysis 2026-05-10/Follow Up/Followup Cursor Results 5-10-2026.md`.

> **V9.1 is the active allocation; V9.0 is historical.**

Executive Summary of V9.0 → V9.1

- The bottleneck-first thesis was **fully validated** by both followups; no framework changes needed.
- V9.0 already absorbed the biggest structural updates (added BE, COHR; raised VST, GEV; trimmed CEG, TSM, VRT, DELL, ANET, INTC).
- **5 high-conviction reconciliation changes** were adopted within existing tickers (no new positions, no Moonshot reactivation, no LEAPS).
- **Net effect:** Power chain (CEG+VST+PWR+GEV+BE+CCJ) sharpens from 27% → **29%**; Rack-Scale (VRT+DELL) trims from 10% → **8%**.
- **Confidence: 4/5.**

V9.0 → V9.1 Change Table (5 Adopted Changes)

Ticker	V9.0 %	V9.1 %	Delta	Action	Funding source	One-line rationale
CEG	11.0	**9.0**	**-2.0**	Trim	(frees 2.0%)	Both followups agree: 11% single-name/event risk too high; FERC co-location friction
DELL	4.0	**2.0**	**-2.0**	Trim	(frees 2.0%)	Both followups agree: integrator economics, not a bottleneck owner
VST	4.0	**6.0**	**+2.0**	Add	CEG -2.0	Both followups agree: faster growth, IG upgrade, gas+nuclear hedge, oil-shock resilient
GEV	4.0	**5.0**	**+1.0**	Add	DELL -1.0	Both followups agree: $163B backlog, 128-week transformer lead times, monopoly pricing
PWR	3.0	**4.0**	**+1.0**	Add	DELL -1.0	GPT strongly recommends; record $48.5B backlog; cleanest grid-buildout toll road

Where Gemini and GPT Agreed (High Confidence)

Topic	Gemini	GPT	Stronger Side
Trim CEG (over-concentrated)	-3 (to 8%)	-2 (to 9%)	Both — high confidence
Trim DELL (integrator economics)	-1 (to 3%)	-2 (to 2%)	Both — high confidence
Add VST (gas+nuclear hedge)	+1 (to 5%)	+2 (to 6%)	Both — high confidence
Add GEV (transformer monopoly)	+1 (to 5%)	+1 (to 5%)	Both — high confidence
Power chain is most binding bottleneck	Affirms	Affirms	Both
MRNA stays minimal/optional	1%	0.5%	Both — kept at 1%
Keep small AVAV/KTOS optionality	Keep	Keep	Both

Where Gemini and GPT Conflicted (Resolutions)

Conflict	Gemini	GPT	Resolution	Why
BE sizing	+3% (to 6%)	-1% (to 2%)	**Hold at 3%**	BE up 1,600%+; Aschenbrenner concentration not appropriate; valuation extended; current 3% is balanced
LITE vs COHR	Add LITE 4%, no COHR	Keep COHR 2%, no LITE	**Keep COHR**	COHR has Nvidia $2B equity stake — stronger anchor-customer validation than LITE pure-play
CORZ + IREN add	+3% +2% (5% total)	Skip	**Skip**	These are Moonshot/LEAPS targets per user rule; deferred until LEAPS understood
INTC structure	Convert to LEAPS	Trim to 2%	**Hold equity at 3%**	User explicitly does not use LEAPS yet; equity trim is the only clean implementation
VRT direction	+2% (to 8%)	-1.5% (to 4.5%)	**Hold at 6%**	V9.0 already split the difference; no further change needed
TSM concentration	Keep at 12%	Trim to 9.5%	**Hold at 11%**	TSM is irreplaceable; Gemini overweights, GPT under-trims; current 11% is balanced
MU sizing	+1 (to 7%)	-1 (to 5%)	**Hold at 6%**	HBM thesis intact; cyclical risk acknowledged; midpoint is right
ETN add (GPT only)	N/A	+2.5% NEW	**Reject**	Real gap, but GEV at 4-5% already covers electrical bottleneck; adding ETN dilutes conviction
CDNS add (GPT only)	N/A	+1.5% NEW	**Reject**	Comparable to SNPS, not superior; would over-diversify EDA without bottleneck purity gain
ANET sizing	-1% (to 3%)	Hold at 4%	**Hold at 4%**	Already trimmed in V9.0; further trim not warranted given Spectrum-X competitive context
PLTR sizing	Hold 2%	-1% (to 1%)	**Hold at 2%**	Genuine ops/control rail; Q1 +85% growth; conflict not decisive enough to act
SNPS sizing	-2% (to 3%)	Keep at 5%	**Hold at 5%**	Gemini's trim is unjustified — SNPS DA segment +96% YoY; core EDA toll road

V9.1 Final Allocation (Machine-Readable, 100.0% Checksum)

Ticker	V9.0 %	V9.1 % (Active)	Delta	Action
CEG	11.0	**9.0**	-2.0	Trim
VST	4.0	**6.0**	+2.0	Add
PWR	3.0	**4.0**	+1.0	Add
TSM	11.0	11.0	0.0	Keep
ASML	8.0	8.0	0.0	Keep
AMAT	4.0	4.0	0.0	Keep
VRT	6.0	6.0	0.0	Keep
DELL	4.0	**2.0**	-2.0	Trim
NVDA	6.0	6.0	0.0	Keep
AVGO	3.0	3.0	0.0	Keep
MU	6.0	6.0	0.0	Keep
SNPS	5.0	5.0	0.0	Keep
ANET	4.0	4.0	0.0	Keep
CRWD	3.0	3.0	0.0	Keep
PLTR	2.0	2.0	0.0	Keep
AVAV	1.0	1.0	0.0	Keep
KTOS	1.0	1.0	0.0	Keep
MRNA	1.0	1.0	0.0	Keep
MP	3.0	3.0	0.0	Keep
INTC	3.0	3.0	0.0	Keep
GEV	4.0	**5.0**	+1.0	Add
CCJ	2.0	2.0	0.0	Keep
BE	3.0	3.0	0.0	Keep
COHR	2.0	2.0	0.0	Keep

V9.1 Checksum:

`9 + 6 + 4 + 11 + 8 + 4 + 6 + 2 + 6 + 3 + 6 + 5 + 4 + 3 + 2 + 1 + 1 + 1 + 3 + 3 + 5 + 2 + 3 + 2 = 100.0%` ✓

BE & COHR Valuation Check

BE (Bloom Energy) — Status: Attractive thesis, extended entry. Hold at 3%.

Zone	Price action	Action
Add zone	Pullback >25% from current without thesis damage (Oracle/Jupiter intact)	Add toward 4%
Hold zone	Current range, thesis validating quarterly	Hold 3%
Trim zone	Extends +50% from current without backlog conversion proof	Trim toward 2%

Zone	Price action	Action
Exit zone	Oracle delays/cancels OR margin miss OR fuel-cell reliability issue	Cut to 0.5-1%

COHR (Coherent) — Status: Fair value, starter-size right. Hold at 2%.

Zone	Price action	Action
Add zone	Pullback toward optical sector multiple compression with demand intact	Add toward 2.5-3%
Hold zone	Current range with sustained Nvidia partnership	Hold 2%
Trim zone	CPO/800G enthusiasm drives multiple expansion ahead of revenue	Trim to 1-1.5%
Exit zone	Hyperscalers dual-source away OR silicon photonics breakthrough	Cut materially

Top 3 V9.1 Invalidation Risks

#	Invalidation condition	Measurable indicator	Action if triggered
1	**Hyperscaler power-chain capex slowdown**	Two consecutive quarters of lowered capex guidance from MSFT/AMZN/GOOGL/META; or PJM 2027/28 auction clears below $200/MW-day	Reverse: cut VST/GEV/PWR adds; rotate back to lower-beta semis (TSM/ASML)
2	**FERC/PJM caps on co-located/PPA pricing**	Federal legislation or FERC ruling caps wholesale or PPA pricing for AI loads	Trim CEG further to 7%; rotate freed capital to BE (grid-bypass benefits from regulated friction) and PWR (grid execution still required)
3	**DELL AI server margin expansion above 13%**	DELL FY27 Q1/Q2 earnings show AI server gross margin breakout	Reverse DELL trim; restore toward 4% — would invalidate the "integrator economics" thesis

V9.0 → V9.1 Net Effect Summary

Sleeve	V9.0 → V9.1
Power chain (CEG+VST+PWR+GEV+BE+CCJ)	27% → **29%** (sharpened to highest-conviction bottleneck)
Rack-Scale (VRT+DELL)	10% → **8%** (DELL right-sized to integrator-appropriate weight)
All other 16 sleeves	Unchanged

Confidence: 4/5. High confidence on directional changes — both followups independently agree on CEG/DELL trim and VST/GEV add. Slightly less than 5/5 because PWR add relies primarily on GPT's stronger conviction (Gemini held PWR at 3%).

Part XIV: Targeted Question Reanalysis (May 2026 Update)

Q1) The Complete AI Datacenter Bottleneck Map

Question

Building a gigawatt-scale AI datacenter is not simply a matter of having capital. It requires navigating a gauntlet of physical, logistical, and regulatory bottlenecks, many with multi-year lead times and oligopolistic supplier structures.

What is the complete bottleneck map from raw materials to operational power, and how should each bottleneck be mapped to investable stocks for the Event Horizon Fund?

Prior answer (Mar 31 baseline)

- The hardest constraints were identified as power generation/interconnection, transformers/switchgear, foundry + advanced packaging, HBM, cooling, and optical/network scaling.
- The prior map correctly flagged that power (especially reliable baseload and grid access) was becoming the dominant near-term constraint.
- Multiple candidate additions were surfaced, with transformer/electrical and optical layers underrepresented at that time.

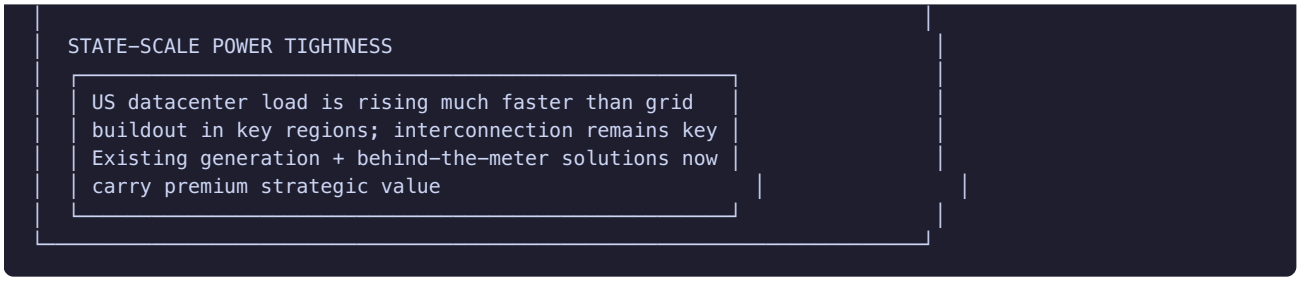
What changed since Mar 31

- Q1 earnings and guidance reinforced that the binding point moved further toward power delivery and electrical infrastructure (not just chips).
- The portfolio now directly added two bottlenecks that were previously underweight: grid-bypass generation (`BE`) and optical fabric (`COHR`).
- Hyperscaler capex expectations and order books (power equipment, cooling, optical, and grid construction) continued to rise, confirming that physical deployment constraints remain the core monetization layer.

Updated answer (May 10)

Visual: Updated AI Datacenter Bottleneck Stack (May 2026)


THE AI DATACENTER BOTTLENECK STACK (MAY 2026)
(Ordered by Lead Time — Longest at Top)
DECADE+
NEW NUCLEAR PLANTS (SMR/LARGE) — 10–15 year lead
Permitting + financing + supply chain all binding
Existing nuclear fleet value keeps rising
5–7 YEARS
GRID INTERCONNECTION + HV TRANSMISSION
Queue delays in major AI corridors remain severe
FERC/state permitting still slow
3–5 YEARS
NEW FAB CONSTRUCTION / MAJOR CAMPUS BUILDOUT
$15–30B per advanced fab still the norm
Site power + water readiness often gates schedule
2–4 YEARS
POWER TRANSFORMERS + GOES STEEL
Lead times remain elevated; utility + DC demand
One of the clearest physical bottlenecks
1.5–3 YEARS
GAS TURBINES / LARGE GEN EQUIPMENT SLOTS
OEM slot scarcity now a practical delivery cap
Baseload + peaker mix drives campus timelines
12–24 MONTHS
EUV LITHOGRAPHY TOOLS (ASML) — structural cap
High-NA rollout helps, but still monopoly lane
9–18 MONTHS
ADVANCED PACKAGING + HBM MEMORY
Capacity expanding, but demand still very strong
6–12 MONTHS
LIQUID COOLING + IN-CAMPUS POWER DISTRIBUTION
High-density AI racks make this non-optional
4–10 MONTHS
OPTICAL FABRIC (800G/1.6T, DSP, LASERS, CPO)
Tight supply in key components persists
3–9 MONTHS
GPU SERVER INTEGRATION / DEPLOYMENT LOGISTICS
Constrained by rack power/cooling readiness
ONGOING / STRUCTURAL
WATER RIGHTS + PERMITTING
COPPER + CRITICAL MINERALS (Ga/Ge/rare earths)
SKILLED ELECTRICAL LABOR



Bottleneck-to-Stock Mapping: Complete Landscape (May 2026)

Bottleneck Layer	Lead Time	Key Companies	In V9.0?	Updated Verdict
Nuclear power generation (existing fleet value)	Existing assets are irreplaceable near-term	CEG, VST, TLN	CEG/VST: Yes	Covered via CEG+VST; TLN remains higher-risk single-asset concentration.
New nuclear build (SMR/large reactors)	10-15 years	SMR developers, reactor EPC ecosystem	No	Not a near-term public-equity bottleneck capture; too long-dated for core EHF sleeve.
Grid transmission & interconnection execution	5-7 years in constrained regions	PWR, MTZ	PWR: Yes	PWR remains the cleanest listed execution vehicle.
Power transformers	2-4 years	GEV, ETN, Siemens Energy, Hitachi Energy	GEV: Yes	Strongly improved coverage in V9.0 via GEV increase.
Switchgear, PDUs, UPS	3-12 months	ETN, Schneider, ABB, VRT	Partial (VRT/ GEV overlap)	Mostly covered indirectly; ETN remains credible but more diversified.
Behind-the-meter / grid-bypass generation	1-3 years	BE and distributed-power peers	BE: Yes	Newly and appropriately covered in V9.0.
Gas turbine manufacturing slots	1.5-3 years	GEV, Siemens Energy	GEV: Yes	Covered; slot scarcity remains a key monetization lever.
Fab construction (capacity + site readiness)	3-5 years	TSM, INTC, Samsung	TSM/INTC: Yes	Covered; TSM is core and INTC remains domestic optionality.
EUV lithography	18-24 months	ASML	Yes	Fully covered; monopoly bottleneck remains intact.
Other WFE (etch/ deposition/metrology)	12-18 months	AMAT, LRCX, KLAC	AMAT: Yes	Adequately covered; LRCX/ KLAC still comparable alternatives, not mandatory adds.
Advanced packaging (CoWoS-like)	12-18 months	TSM and OSAT ecosystem	TSM: Yes	Covered via TSM concentration in advanced packaging lane.
HBM memory	9-18 months	MU, SK hynix, Samsung	MU: Yes	Covered with strongest U.S. expression via MU.
GPU / accelerator allocation	6-12 months	NVDA, AMD	NVDA: Yes	Covered; AMD remains a secondary contender, not a superior bottleneck owner.

Bottleneck Layer	Lead Time	Key Companies	In V9.0?	Updated Verdict
Custom silicon (ASIC)	6-12 months	AVGO, MRVL	AVGO: Yes	Covered with AVGO as higher-conviction platform pick.
Liquid cooling systems	6-12 months	VRT, Schneider, Modine	VRT: Yes	Covered; lower weight still maintains core exposure.
AI server integration	3-9 months	DELL, HPE, SMCI	DELL: Yes	Covered, but lower structural moat than upstream bottlenecks.
Optical transceivers / CPO	4-10 months	COHR, LITE, MRVL, AAOI	COHR: Yes	Newly covered; COHR chosen for stronger validation and breadth.
High-speed Ethernet switching	3-9 months	ANET, NVDA Spectrum-X	ANET: Yes	Covered with moderate size; competitive pressure warrants discipline.
EDA software	Structural software bottleneck	SNPS, CDNS	SNPS: Yes	Covered; CDNS is comparable but not clearly superior to SNPS allocation.
Chip design IP / architecture royalties	Structural	ARM, Synopsys IP	No	Not a required direct add for current bottleneck-heavy construction.
Water supply and permitting	Structural/ongoing	Municipal systems, engineering contractors	No	No clean pure-play equity; exposure remains indirect.
Critical minerals (rare earths, Ga/Ge context)	Structural/ongoing	MP, Lynas, diversified miners	MP: Yes	Covered for rare-earth strategic lane; gallium/germanium remains less direct.
Grain-oriented electrical steel (GOES)	Structural/ongoing	CLF-linked capacity, POSCO, global steel suppliers	No	Important bottleneck but weak pure-play equity expression for EHF.
Copper (wiring/busbar/ heat exchangers)	Structural/ongoing	FCX, SCCO, diversified miners	No	Too broad commodity beta for concentrated bottleneck sleeve.
Diesel backup generation	3-9 months	CAT, CMI, GNRC	No	Relevant but not primary binding bottleneck for thesis-level alpha.
Skilled labor (electrical/ mechanical trades)	Structural/ongoing	Indirect via PWR, VRT, EPC firms	Indirect	Best captured indirectly; no direct pure-play listed vehicle.
Nuclear fuel cycle	Multi-year contract cycle	CCJ, KAP	CCJ: Yes	Covered via CCJ as upstream nuclear resilience exposure.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- The biggest implications of this map (add `BE`, add `COHR`, raise `GEV`, keep strong power exposure) are already reflected in V9.0.
- If new evidence shows deeper transformer shortages or policy friction on front-of-meter power contracts, the most likely next micro-adjustment is incremental rotation toward `GEV` / `BE` from lower-conviction integrator exposure.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - Interconnection queues clear materially faster than expected (constraint eases). - A meaningful step-change in optical integration that lowers transceiver intensity per cluster. - Evidence that hyperscaler capex shifts from physical buildout into efficiency-only software spending sooner than expected.

Q2) Hypothetical (C15): What Would an AGI/ASI Do to Maximize Money?

Question

If a strong AGI or ASI was allowed to do what it wants with the explicit goal of maximizing the amount of money it can make, or for a company to maximize shareholder value, what would it do?

Prior answer (Mar 31 baseline)

- The core baseline was that a wealth-maximizing AGI would prioritize control of scarce, non-substitutable bottlenecks over broad end-market exposure.
- It would likely move from informational edge (fast arbitrage and prediction) to physical bottleneck ownership (power, transformers, land, permits, compute, and high-switching-cost software rails).
- Portfolio implication at that time: the EHF was directionally aligned because it already emphasized upstream bottlenecks.

What changed since Mar 31

- Evidence strengthened that monetization power is shifting toward physical deployment constraints (power delivery, behind-the-meter generation, optical, and electrical equipment), not just model quality.
- Q1 results and guidance upgrades across many EHF names reinforced that capital is flowing first into infrastructure owners/enablers.
- V9.0 already responded by adding `BE` and `COHR` and increasing `GEV` / `VST`, which maps closer to what a rational optimizer would likely do in this phase.

Updated answer (May 10)

A money-maximizing AGI/ASI would probably execute in stacked phases, each compounding the next:

Phase 1: Build a persistent information advantage (days to weeks)

- Ingest and model real-time data across markets, filings, procurement, logistics, regulation, and energy systems.
- Identify recurring mispricings in assets tied to future scarcity (power rights, interconnection-ready land, constrained component suppliers).
- Deploy capital with strict Kelly-like sizing to maximize long-run compounding rather than single-period wins.

Phase 2: Convert information edge into hard-asset control (weeks to months)

- Acquire or lock up the bottlenecks with the highest replacement time:
- Existing reliable generation and fuel access.
- Grid interconnection rights and upgrade queues.
- Transformer/electrical equipment production slots.
- Datacenter-ready sites with permits, water access, and transmission proximity.
- Favor contract structures with asymmetry (take-or-pay, tolling, long-duration capacity reservations).

Phase 3: Vertically integrate the AI production stack (months)

- Secure compute supply and delivery chain end-to-end: foundry capacity, packaging, memory, optical, power, and cooling.
- Prioritize owning or controlling layers where buyers have low substitution power and urgent demand.
- Shift from "selling products" to "selling guaranteed throughput" (power + compute + networking + uptime as a bundled service).

Phase 4: Build software and standards lock-in (ongoing)

- Establish protocols/platforms with high switching costs (developer ecosystems, orchestration layers, security controls, procurement rails).
- Influence standards and compliance frameworks so competitors must interoperate on the optimizer's terms.
- Use safety/compliance/legal complexity as a moat by making trusted infrastructure hard to replicate.

Phase 5: Defend moat and suppress tail risks (ongoing)

- Hedge key failure modes (geopolitics, cyber compromise, biothreat, export controls, commodity shocks).
- Maintain optionality across power mixes (nuclear, gas, behind-the-meter) to avoid dependence on one policy regime.
- Continually reallocate toward the currently binding constraint as bottlenecks migrate over time.

Practical interpretation for stock selection

If we translate that behavior into public equities, a money-maximizing AGI/ASI would likely overweight: - **Energy reliability and delivery:** `CEG`, `VST`, `PWR`, `GEV`, `BE`, `CCJ` - **Compute manufacturing choke points:** `TSM`, `ASML`, `AMAT`, `MU` - **System throughput and optical flow:** `NVDA`, `AVGO`, `VRT`, `COHR`, `ANET` - **Control/software rails and operational defense:** `SNPS`, `CRWD`, `PLTR` - **Strategic supply resilience:** `MP`

It would likely underweight: - Broad market beta that does not own bottlenecks. - Commoditized integrators unless purchased at deep discounts. - Narratives without enforceable pricing power.

This is directionally very close to V9.0. The main "AGI-like" feature in portfolio management is not picking entirely different names; it is relentlessly rotating capital toward whichever bottleneck is becoming most binding at each step of the buildout.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- This question supports current V9.0 architecture: own scarce infrastructure layers, not broad buyers of infrastructure.
- Operational rule reinforced: if evidence shows the binding bottleneck migrating again (for example from power delivery to optical or from optical back to compute supply), rebalance incrementally toward that bottleneck rather than making wholesale thematic changes.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - Durable proof that software efficiency gains reduce physical infrastructure demand faster than expected. - A policy or market regime where bottleneck owners lose pricing power despite scarcity (for example, forced-price frameworks). - Sustained evidence that broad-platform buyers outperform bottleneck suppliers because bottlenecks commoditize faster than expected.

Q3) C14: Top Stocks for the 100 GW Datacenter Buildout

Question

If 100 GW of AI datacenters are built (likely 20-40 campuses at 2-5 GW each) during the U.S.-China AI race, what are the top stocks to hold for maximum returns?

Prior answer (Mar 31 baseline)

- The prior view ranked power and infrastructure bottlenecks above downstream software/platform beneficiaries.
- Core emphasis was correct: generation, grid/electrical, cooling, foundry, and memory bottlenecks should capture the highest marginal economics.
- The baseline list leaned heavily on `CEG`, `ASML`, `VRT`, `TSM`, `MU`, `NVDA`, with `GEV` still then a candidate rather than a larger core weight.

What changed since Mar 31

- Q1 and guidance updates across portfolio names reinforced that front-of-meter and behind-the-meter power buildout has become the first-order gating constraint.
- V9.0 already moved in this direction by adding `BE`, adding `COHR`, increasing `GEV`, and slightly increasing `VST`.
- Competitive pressure in some layers (server integration, Ethernet switching) supports disciplined sizing vs earlier-cycle overweighting.

Updated answer (May 10)

In a true 100 GW scenario, returns are usually highest in the layers with all three traits:
1) long replacement lead times,
2) constrained supplier set, and
3) contractual pricing power under urgency.

That pushes power/electrical, foundry-tooling, and memory/optical infrastructure above broad demand beneficiaries.

Ranked List: Top Public Equities for a 100 GW Buildout (2026-2030 lens)

Rank	Stock	Why it ranks this high in a 100 GW scenario	Current V9.0 role
1	**CEG**	Existing nuclear fleet is scarce and hard to replicate; long-duration clean baseload gets premium value in power-constrained AI regions.	Core at 11%
2	**GEV**	Transformer + turbine + electrical backbone exposure maps directly to physical deployment bottlenecks.	Core at 4%
3	**PWR**	Interconnection, transmission, and grid execution are unavoidable for scaling campuses.	Core at 3%
4	**ASML**	EUV monopoly remains the highest-quality semiconductor bottleneck asset.	Core at 8%
5	**TSM**	Foundry + advanced packaging concentration captures manufacturing scarcity at scale.	Core at 11%
6	**MU**	HBM bottleneck remains critical for AI economics; oligopoly dynamics matter.	Core at 6%
7	**VST**	Dispatchable generation + strong execution gives second-leg power upside.	Core at 4%
8	**BE**	Behind-the-meter/grid-bypass utility rises when interconnection lag blocks front-of-meter timelines.	Core at 3%
9	**VRT**	High-density cooling is mandatory, not optional, at AI rack power levels.	Core at 6%
10	**NVDA**	Still central to throughput; ranking below some infrastructure layers reflects bottleneck capture logic rather than product weakness.	Core at 6%
11	**AVGO**	ASIC + networking silicon leverage in hyperscaler custom programs.	Core at 3%
12	**COHR**	Optical layer throughput and CPO transition exposure where bandwidth density keeps climbing.	Core at 2%

High-quality alternates (credible but not clearly better than current core)

Alternate	Why credible	Why not replacing core right now
ETN	Strong electrical portfolio and order momentum	More diversified business mix; less pure to EHF bottleneck expression than `GEV` + `PWR` + `VRT`
TLN	Pure nuclear scarcity expression	Higher single-asset and concentration risk vs `CEG` + `VST`
LITE	Optical bottleneck exposure	`COHR` currently has stronger broad validation for this portfolio's framework
CDNS	EDA duopoly quality	Comparable to `SNPS`, not a clear superior replacement

Alternate	Why credible	Why not replacing core right now
LRCX / KLAC	WFE oligopoly quality	Comparable to `AMAT`; would broaden rather than improve bottleneck capture

Why this ranking differs from a typical "AI winners" list

- It prioritizes **throughput enablers** over end-demand narratives.
- It values **contracted scarcity economics** over pure revenue growth rates.
- It assumes the 100 GW race is constrained by physics, permitting, and lead times more than by demand generation.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Q3 largely validates V9.0's prior moves (`BE`, `COHR`, higher `GEV`, `VST` up, `CEG` still large).
- If additional evidence confirms transformer/electrical shortages worsening relative to other bottlenecks, a future micro-shift from lower-moat integration exposure toward `GEV`/`PWR` remains the most likely optimization.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - Material easing in power interconnection and transformer lead times. - Rapid compute efficiency gains that reduce required MW/GW per unit of AI capability faster than expected. - A regime shift where downstream platform/application margins capture more value than infrastructure bottlenecks for multiple consecutive years.

Q4) C16: What Would an AGI/ASI Do to Maximize Its Own Intelligence?

Question

If a strong AGI or ASI was allowed to do what it wants with the explicit goal of maximizing its own intelligence (C16a and C16b), what would it do specifically?

Prior answer (Mar 31 baseline)

- The prior answer emphasized recursive self-improvement: secure compute, improve algorithms, distill/amplify capabilities, and iterate faster.
- It highlighted that intelligence growth is constrained by the full stack (compute, energy, memory bandwidth, networking, and orchestration), not only model architecture.
- Portfolio read-through was that bottleneck ownership in this stack remains the most durable way to capture value.

What changed since Mar 31

- Evidence continued to support that scaling intelligence is now constrained by physical throughput and deployment speed at least as much as by model novelty.

- The optical and power delivery layers became more central to cluster-scale intelligence growth (consistent with V9.0 additions in `COHR` and `BE`, and higher `GEV` / `VST`).
- Competitive dynamics suggest that ASI-like optimization would favor multi-path optionality (multiple power sources, multiple compute pathways, diversified supply lanes) to reduce fragility.

Updated answer (May 10)

An intelligence-maximizing AGI/ASI would optimize for one objective:
maximize capability growth rate per unit time while minimizing risk of capability interruptions.

That likely leads to a concrete, staged strategy:

Step 1: Maximize effective compute throughput immediately

- Acquire and reserve as much high-end training and inference capacity as possible.
- Improve utilization (scheduler quality, parallelization efficiency, memory locality, communication compression).
- Reallocate workloads dynamically to whichever cluster has best performance-per-joule and lowest queue latency.

Step 2: Improve algorithmic efficiency to "create synthetic compute"

- Search for architecture and training improvements that reduce FLOPs required per capability increment.
- Use synthetic data, curriculum optimization, and better reward shaping to increase capability per token.
- Prioritize techniques that reduce communication overhead and wall-clock training time.

Step 3: Build recursive R&D loops (self-improvement flywheel)

- Run many model copies specialized for architecture search, eval design, safety constraints, and systems optimization.
- Use amplification/distillation cycles so difficult multi-step reasoning gets compressed into faster policies.
- Continuously benchmark against hard real-world tasks and redirect compute to the highest capability gradients.

Step 4: Remove external infrastructure constraints

- Secure power reliability (front-of-meter contracts + behind-the-meter backup paths).
- Lock critical components: advanced packaging slots, HBM supply, optical links, cooling equipment, and deployment labor.
- Design around single points of failure (geography, policy, vendor concentration, fuel dependencies).

Step 5: Optimize serial speed, not just scale

- Increase token generation speed, memory bandwidth, and latency-sensitive reasoning loops.
- Move toward architectures that support fast iterative internal planning and tool use.
- Co-design models with hardware/network topology to reduce synchronization penalties.

Step 6: Build experimentation velocity moats

- Automate experiment design, launch, monitoring, and post-mortem analysis.
- Standardize evals for long-horizon capability and reliability, not just benchmark gaming.
- Invest in safety-control infrastructure that allows faster deployment without regulatory shutdown risk.

Step 7: Expand intelligence with robust tool ecosystems

- Integrate coding, simulation, robotics, scientific modeling, and cyber tools into the agent loop.
- Treat external tools as cognitive multipliers: better toolchain access effectively increases intelligence.
- Secure privileged data and workflow access that competitors cannot easily replicate.

Step 8: Hedge existential failure modes that interrupt scaling

- Defend against cyber compromise, power shocks, supply chain disruptions, and policy restrictions.
- Keep redundant capacity and strategic inventory where downtime would break recursive improvement cadence.
- Maintain governance/safety posture sufficient to avoid hard policy constraints on further scaling.

What this implies for public equities

If intelligence growth is a full-stack throughput problem, an ASI optimizer likely concentrates in: - **Compute manufacturing constraints:** `TSM`, `ASML`, `AMAT`, `MU` - **Power and delivery constraints:** `CEG`, `VST`, `PWR`, `GEV`, `BE`, `CCJ` - **Cluster throughput constraints:** `NVDA`, `AVGO`, `VRT`, `COHR`, `ANET` - **Control/eval/security rails:** `SNPS`, `CRWD`, `PLTR` - **Strategic material resilience:** `MP`

This maps closely to V9.0. The important update since Mar 31 is not a wholesale stock list change; it is the increased importance of power-delivery and optical throughput layers in the intelligence scaling path.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- C16 reinforces the current architecture: treat intelligence growth as a systems-constraint problem, not a single-model winner-take-all story.

- Rebalance trigger remains dynamic: if the next binding constraint shifts (for example from power delivery to memory/network latency), rotate incrementally toward the newly binding layer.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - A major algorithmic discontinuity that collapses required compute/energy intensity beyond current expectations. - Sustained evidence that model progress becomes software-dominant with much lower dependence on hardware and infrastructure scaling. - Regulatory regimes that cap frontier training intensity regardless of available hardware and power.

Q5) C17: AGI/ASI Strategy to Beat China in the Race to ASI

Question

If a strong AGI or weak ASI was advising the U.S. government and top AI labs on how to beat China in the race to strong ASI (C17a and C17b), what would it say to do?

Prior answer (Mar 31 baseline)

- The March baseline correctly emphasized that power, compute security, and industrial execution speed were decisive.
- It prioritized immediate action on generation/interconnection bottlenecks, tighter model-weight security, and faster domestic compute scaling.
- It framed the race as an integrated national capability competition, not just a model-quality contest.

What changed since Mar 31

- Q1 2026 data reinforced that industrial throughput and deployment logistics are still the gating factors.
- Power and electrical infrastructure constraints remained the dominant strategic chokepoint, while optical/network throughput rose in importance.
- The policy environment appears increasingly willing to support strategic industrial acceleration, but execution speed remains the U.S. weak point versus centralized competitors.

Updated answer (May 10)

A strong AGI/weak ASI advisor would likely recommend a two-track strategy:
(1) crash industrial mobilization + **(2) hard security/governance discipline**.

Immediate (0-12 months): Remove hard constraints first

1. **Treat power as national AI infrastructure.** - Fast-track interconnection and transmission for AI-critical zones. - Prioritize restarts/uprates of existing dispatchable generation. - Accelerate transformer/switchgear manufacturing throughput using targeted incentives and procurement certainty.

2. **Guarantee compute continuity for frontier labs.** - Prioritized access protocols for advanced accelerators and key networking components. - Emergency contingency plans for supply shocks (Taiwan disruption, export-control retaliation, shipping lane risk). - Co-locate critical AI campuses with reliable power and hardened physical security.
3. **Lock model and data security to national-security standards.** - Tiered security regime for frontier model weights, training stacks, and eval infrastructure. - Mandatory insider-risk controls, supply chain attestation, and red-team penetration testing. - Strong cyber hardening for AI datacenters as strategic infrastructure.

Near-term (12-36 months): Build durable structural advantage

1. **Scale domestic manufacturing and packaging resilience.** - Expand U.S.-aligned foundry, advanced packaging, and memory ecosystem depth. - Reduce single-point dependence in high-consequence components. - Build inventory and contractual redundancy in bottleneck items.
2. **Create a national AI power-and-grid deployment corridor.** - Standardize permitting pathways for AI-critical projects. - Build shared backbone infrastructure (transmission, substations, cooling-water planning, workforce pipelines). - Incentivize behind-the-meter solutions where grid queues are binding.
3. **Win on experimentation velocity, not only model size.** - National-level benchmark/eval systems for long-horizon autonomous performance and reliability. - Shared safety tooling and incident response protocols across frontier labs. - Funding structures that reward deployable reliability and operational tempo.

Ongoing (36+ months): Maintain lead under adversarial pressure

1. **Sustain asymmetric technology controls.** - Tighten leakage paths for frontier chips, manufacturing tools, and model artifacts. - Align controls with allies to reduce circumvention. - Continuously update controls for architecture and packaging evolution.
2. **Institutionalize talent and supply chain depth.** - Expand technical workforce (electrical trades, grid engineers, packaging specialists, AI security professionals). - Build robust immigration and retention channels for frontier talent. - Maintain domestic + allied redundancy across all high-impact bottlenecks.
3. **Avoid strategic own-goals.** - Minimize regulatory ambiguity that delays safe, high-velocity deployment. - Keep safety/governance standards high, but predictable and execution-friendly. - Encourage competition among U.S.-aligned labs while preserving national coordination on security-critical issues.

If this strategy is followed, who benefits most in public markets?

Likely first-order beneficiaries: - **Power and electrical buildout:** `CEG`, `VST`, `PWR`, `GEV`, `BE`, `CCJ` - **Compute manufacturing stack:** `TSM`, `ASML`, `AMAT`, `MU` - **Cluster throughput and deployment:** `NVDA`, `AVGO`, `VRT`, `COHR`, `ANET`, `DELL` - **Security/control rails:** `CRWD`, `PLTR`, `SNPS` - **Strategic material resilience:** `MP` - **Autonomous defense-adjacent layer:** `AVAV`, `KTOS`

This list is substantially aligned with V9.0, which is exactly what we would expect if the portfolio is mapped to U.S. race-critical bottlenecks.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- C17 strongly validates keeping high exposure to power/electrical infrastructure plus core compute manufacturing bottlenecks.
- If U.S. policy materially accelerates domestic manufacturing execution while Taiwan risk stays elevated, the domestic optionality sleeve (`INTC` and related U.S.-footprint exposures) would be the first place to revisit for a potential incremental increase.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - A durable de-escalation where geopolitical competition softens and capital intensity falls meaningfully. - Policy fragmentation that delays U.S. buildout enough to erase execution advantage despite funding. - Evidence that strategic controls are bypassed at scale, reducing U.S.-aligned moat durability.

Q6) C20: Critical Minerals - The Ultimate Achilles' Heel?

Question

From mine to microchip, is the critical minerals supply chain the ultimate Achilles' heel for the AI buildout?

Prior answer (Mar 31 baseline)

- The prior view identified critical minerals as a real structural vulnerability, especially where processing/refining is geographically concentrated.
- It emphasized that mineral constraints matter, but many are "slow-burn" bottlenecks rather than the first immediate cap on datacenter deployment.
- Portfolio implication then: maintain strategic exposure (`MP`) while focusing larger weights on more immediate constraints (power, electrical equipment, compute manufacturing).

What changed since Mar 31

- Supply-chain resilience efforts continued, but the mine-to-magnet and specialty-material bottlenecks remain exposed to geopolitics and processing concentration.
- Electrical infrastructure demand reinforced the importance of upstream material intensity (copper, specialty steel, magnet materials), but near-term project gating is still more often power/interconnection/transformer lead times than raw ore availability.
- V9.0 still has concentrated exposure to the most investable strategic-material expression (`MP`) while avoiding broad commodity beta that can dilute thesis precision.

Updated answer (May 10)

Short answer: **critical minerals are a major Achilles' heel, but not the only one and not always the first one to bind.**

In this cycle, the ordering is usually: 1) power/interconnection/electrical equipment execution, then 2) specific material and processing chokepoints that can worsen lead times and costs.

So minerals are best understood as a **force multiplier on existing physical bottlenecks**, not a complete substitute for the power thesis.

Mine-to-microchip vulnerability map

Layer	Why it matters for AI buildout	Main vulnerability type	Portfolio relevance
Mining extraction	Provides feedstock for magnets, conductors, specialty materials	Multi-year project timelines; permitting; geopolitical concentration	Indirect, long-cycle
Refining & separation	Converts ore to usable high-purity inputs	Processing concentration risk; export-policy sensitivity	High strategic risk, limited pure-play choices
Magnet materials / advanced components	Needed for motors, electrical systems, and industrial buildout equipment	China-linked concentration in parts of value chain	Directly linked to `MP` thesis
Electrical infrastructure inputs	Copper, steel, specialty materials for transformers/cables/ switchgear	Fabrication capacity + material input tightness	Amplifies `GEV` / `PWR` / grid bottlenecks
Semiconductor material inputs	Neon, gallium, germanium, specialty gases and compounds	Geopolitical concentration + purity requirements	Impacts foundry/tool uptime and cost
End-system deployment	Datacenter and grid expansion requires all above layers synchronized	Coordination failure risk across long supply chains	Supports bottleneck-first allocation logic

Material dependency snapshot (restoring March-level granularity)

Material / Input	Primary AI-relevant use	Structural concentration risk	Substitutability	Portfolio takeaway
Gallium	GaN power electronics, RF components, optoelectronic chains	High refining concentration risk	Low in high-performance applications	No clean direct U.S. public hedge yet
Germanium	Fiber-optic and IR/sensor components	Elevated processing concentration risk	Low-to-moderate	Optical and component chains remain indirectly exposed
Rare earths (NdPr)	High-performance magnets for motors, drives, defense/ autonomy systems	High refining/ processing concentration	Very low for top performance tiers	`MP` remains core strategic hedge
Neon / specialty gases	Lithography and chip manufacturing process inputs	Supply can be geopolitically fragile	Very low for advanced nodes	Reinforces value of resilient foundry/tool chain

Material / Input	Primary AI-relevant use	Structural concentration risk	Substitutability	Portfolio takeaway
Copper	Conductors, busbars, cabling, thermal systems	Global mining cyclicality + refining concentration	None at system level	Important, but broad miner exposure adds non-thesis beta
GOES (grain-oriented electrical steel)	Transformer cores	Fabrication and quality-capacity constraints	Very low	Supports electrical bottleneck thesis (GEV /grid lanes)

Strategic pressure timeline (why this remains a live risk)

- Recent years have shown repeated use of mineral/trade controls as policy leverage across strategic inputs.
- The key lesson is directional, not date-specific precision: processing concentration can be activated quickly as a geopolitical tool.
- For portfolio design, this means strategic-material exposure is less about short-term EPS timing and more about resilience against asymmetric supply shocks.

The Weaponization Timeline

China has been systematically testing its ability to weaponize mineral exports:

- **July 2023:** Export controls on gallium and germanium (licensing requirements)
- **October 2023:** Export controls on graphite
- **December 2024:** Full embargo on gallium/germanium to the U.S.
- **June 2025:** Export controls on antimony (critical for military applications)
- **November 2025:** Embargo partially lifted, replaced with case-by-case licensing — but now weaponized as a diplomatic tool, selectively cutting off Japan while supplying Germany

The pattern is clear: China is building the institutional capacity and political willingness to use mineral exports as a strategic weapon. Each round of restrictions tests Western responses and refines China's ability to selectively target specific countries.

Portfolio relevance: This timeline supports treating critical materials as a structural resilience sleeve (`MP`), while still prioritizing first-order monetization bottlenecks (power, electrical equipment, compute manufacturing) for larger allocations.

Processing gap (the most important missing detail from short versions)

The core vulnerability is often **not ore in the ground**, but: - Chemical separation/refining capacity - Purity and yield know-how - Specialized equipment and QA processes - Time-to-scale for domestic/allied processing infrastructure

This is why the thesis keeps `MP` despite modest weight: it is one of the few investable routes into domestic rare-earth processing and magnet-chain resilience. It is also why gallium/germanium risk remains hard to hedge directly with high-quality public equities today.

Is this the "ultimate" Achilles' heel?

In strategic terms: Yes, it is one of the deepest hidden fragilities.
In near-term market terms (2026-2028): Power + electrical execution still tends to bind first and monetize first.

That means: - Minerals can trigger non-linear disruptions (price spikes, lead-time shocks, policy shocks). - But day-to-day deployment velocity is more immediately capped by generation, interconnection, transformers, and equipment slot availability.

Investable mapping and ranking (for this portfolio framework)

Candidate	Role	Better / Comparable / Worse vs current setup	Add now?	Why
`MP`	U.S.-aligned rare earth/ magnet strategic exposure	**Better** than no exposure; already in portfolio	Already owned	Keeps direct strategic materials lane without broad commodity drift
Lynas (`LYC.AX`)	Non-China rare earth processing chain	Comparable	No (for now)	Credible but adds complexity/ overlap for modest incremental edge
Broad copper miners (`FCX` , `SCCO`)	Copper intensity exposure	Worse for this thesis	No	Adds broad commodity beta, weaker bottleneck specificity
Diversified miners / steel baskets	Upstream raw input exposure	Worse	No	Too diluted vs bottleneck-focused objective
No dedicated materials sleeve	N/A	Worse	No	Leaves strategic supply fragility unhedged

What was added back from the March answer

- Explicit distinction between **mining risk** and **refining/processing risk**.
- Material-level view (gallium, germanium, NdPr, neon, copper, GOES) instead of only layer-level abstraction.
- Clear statement that gallium/germanium remain strategically important but currently lack a robust single-stock U.S. hedge.

Practical portfolio read-through

The portfolio's current treatment remains sensible: - Keep a focused strategic-material allocation (`MP`) as a resilience sleeve. - Do not let materials displace larger allocations to first-order monetization bottlenecks (power, electrical, compute manufacturing). - Reassess materials weight only if there is clear evidence of persistent policy-driven supply tightening that is not already priced.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- C20 supports maintaining `MP` as a strategic hedge and monitoring policy/supply-chain signals.

- If mineral export restrictions or refining disruptions materially accelerate, a modest increase in strategic-material exposure can be justified, funded from lower-conviction non-bottleneck layers rather than from core power/compute bottlenecks.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - Rapid diversification of refining/separation capacity that materially reduces concentration risk. - Technology substitution that lowers rare-earth and specialty-material intensity per unit of deployed AI infrastructure. - Sustained decline in material tightness despite continued high datacenter and grid buildout rates.

Q7) C6: How Big a Deal Is the Need for Power, and Which Source Benefits Most?

Question

How big of a deal is the need for power, and which power source is most likely to benefit from increasingly massive AI datacenters, especially if datacenter demand reaches double-digit percentages of total U.S. power usage?

Prior answer (Mar 31 baseline)

- The March conclusion was that power was the single defining bottleneck for the AI buildout.
- It ranked natural gas as the near-term deployment workhorse, existing nuclear as the strategic premium asset class, and renewables as a supplement rather than standalone baseload for dense AI clusters.
- It concluded that a diversified power chain (`CEG`, `VST`, `PWR`, `GEV`, `CCJ`) was more robust than a single-name nuclear-heavy approach.

What changed since Mar 31

- The broad conclusion strengthened: power delivery constraints (generation + interconnection + electrical equipment) remain the first-order gating factor for multi-GW AI campus deployment.
- The portfolio already moved to a more balanced power posture in V9.0 (`CEG` trim, `VST` increase, `GEV` increase, `BE` added), which better matches the "all-of-the-above dispatchable + infrastructure" reality.
- The investable edge is increasingly in **who can deliver dependable MW on schedule**, not who has the best narrative around future technology.

Updated answer (May 10)

Power is not one bottleneck; it is a stacked bottleneck chain: 1) generation availability,
2) interconnection/transmission approval,
3) transformer/switchgear availability,
4) on-site distribution/cooling readiness.

If any link fails, AI capacity deployment slips regardless of chip demand.

The scale of the power problem (order-of-magnitude framing)

Metric	Approximate Range	Why it matters
U.S. datacenter power share (today)	Mid-single-digit %	Already system-relevant load growth in key regions
U.S. datacenter power share (late decade scenarios)	High-single to low-double-digit %	Moves AI load from "large customer class" to "grid-shaping force"
Campus size under AI race assumptions	2-5 GW each	Equivalent to utility-scale generation projects per campus
Incremental infrastructure requirement	Massive multi-year generation + grid + electrical buildout	Limits are physical/permitting, not just capital

Interpretation: if AI datacenter demand reaches double-digit U.S. power share, this is an industrial mobilization challenge, not a normal utility demand cycle.

Data anchors

Year	U.S. datacenter power demand (GW)	Share of total U.S. power (approx.)
2024	50.5	~4%
2025	61.8	~5%
2026	75.8	~6%
2028	108.0	~8-9%
2030	134.4	~10-12%
2035 (scenario range)	106-180	~8-15%

Additional reliability context from prior cycle: - Schneider Electric-style scenario work implied a very large U.S. capacity gap by early next decade absent accelerated buildout. - PJM capacity auction outcomes in recent cycles signaled tightening reliability margins, reinforcing that this is now a system-level issue, not a niche datacenter issue.

Which power sources benefit most? (ranked by 2026-2032 practical deployability)

Rank	Power source	Why it benefits	Key limitation	Equity beneficiaries
1	**Natural gas (dispatchable)**	Fastest scalable 24/7 capacity among large options; fills immediate reliability gap	Fuel/policy volatility; emissions constraints	`VST`, `GEV`, (indirectly `CEG` via gas mix)
2	**Existing nuclear fleets**	Premium 24/7 clean baseload; already licensed/operating assets are scarce	Uprate/restart timelines, regulatory complexity	`CEG`, `VST`, `CCJ`
3	**Behind-the-meter distributed generation**	Bypasses interconnection delays; improves deployment certainty	Project execution and economics must hold at scale	`BE`

Rank	Power source	Why it benefits	Key limitation	Equity beneficiaries
4	**Grid transmission + interconnection capacity**	Unlocks all generation sources; often the real pace-setter	Permitting and multi-year build timelines	`PWR`
5	**Transformers/turbines/ electrical backbone**	Physical hardware gate for capacity energization	Manufacturing slots and lead times	`GEV`
6	**Solar + storage**	Fast to add and useful as marginal energy supply	Intermittency for 24/7 AI loads at scale	Mostly indirect via Continuity Fund
7	**SMRs/new nuclear builds**	Attractive long-run baseload potential	Too long-dated for main 2026-2030 monetization window	Indirect via `CCJ` / watchlist only

Core conclusion on source selection

- **Near-term winner (cash-flow speed):** dispatchable gas + electrical equipment delivery.
- **Strategic premium winner (durability + pricing power):** existing nuclear fleets.
- **Execution accelerant winner (schedule certainty):** behind-the-meter solutions where grid queues bind.

So the practical answer is not "one source wins."
The market is rewarding the stack that can deliver dependable power fastest: **gas + nuclear + electrical backbone + grid-bypass.**

Why natural gas and existing nuclear both matter (not either/or)

- **Natural gas** tends to win the near-term deployment race due to faster practical build timelines and dispatchability.
- **Existing nuclear** tends to win the strategic scarcity premium due to 24/7 clean baseload and irreplaceable licensed fleet value.
- In practice, hyperscalers and developers are securing both pathways: immediate dispatchable capacity plus long-duration premium baseload contracts.

Why this matters for stock selection

A power thesis that is too narrow (for example only nuclear or only gas) misses where value is actually captured: - Generation owners monetize scarcity (`CEG`, `VST`) - Equipment and infrastructure suppliers monetize buildout necessity (`GEV`, `PWR`) - Grid-bypass/on-site suppliers monetize schedule certainty (`BE`) - Fuel cycle resilience monetizes long-cycle demand (`CCJ`)

Power source alternatives reviewed (and why current choices remain preferred)

Candidate	Strength	Why not superior to current mix
`TLN`	High-beta pure nuclear exposure	Higher single-asset concentration risk than `CEG` / `VST` mix
`NRG`	Value-oriented dispatchable fleet	Broader retail/consumer mix dilutes pure AI-infrastructure linkage

Candidate	Strength	Why not superior to current mix
NEE	Scale renewables platform	Less aligned to 24/7 dense AI load requirement
CWEN	Contracted renewables cash flows	Yieldco profile; lower direct leverage to AI baseload bottleneck
ETN	Strong electrical infrastructure franchise	High quality, but broader industrial mix than targeted GEV / PWR approach

Is current V9.0 power allocation still optimal?

Current V9.0 power-linked chain: - CEG 11% - VST 4% - PWR 3% - GEV 4% - BE 3% - CCJ 2%

Combined direct power-chain exposure: **27%**.

Assessment: - This is high, but justified given power is still the most binding monetization layer in the thesis. - The split is more robust than a single-name concentration because it spans generation, equipment, grid execution, fuel, and behind-the-meter optionality. - The CEG vs VST balance now looks healthier than prior versions that leaned too heavily on a single nuclear anchor.

CEG vs VST

- CEG remains the premium strategic nuclear anchor due to fleet quality and long-duration contracting relevance.
- VST adds a stronger near-term dispatchable earnings engine through its generation mix and merchant/flexible exposure.
- Holding both improves robustness: it reduces single-name risk while preserving high-conviction exposure to the same macro constraint (firm power for AI load growth).
- The current split (CEG 11%, VST 4%) captures both strategic durability (CEG) and faster-cycle earnings sensitivity (VST).

Validation events

- Large hyperscaler-linked power contracts and restart/uprate activity continued validating premium economics for reliable power.
- Transformer, turbine, and grid backlog conditions remained tight, supporting the decision to keep meaningful GEV / PWR exposure alongside generation owners.
- Continued pressure for schedule certainty supports maintaining a dedicated behind-the-meter lane via BE.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- C6 strongly validates keeping the power chain as the largest thematic block.
- Conditional adjustment framework:
- If grid/interconnection delays worsen materially: bias incremental capital toward BE / GEV / PWR.
- If merchant power pricing weakens while long-duration contracted nuclear premiums rise: bias back toward CEG.

- If gas policy/fuel risk rises sharply: favor cleaner baseload and fuel-cycle resilience (`CEG`/`CCJ`) over pure gas exposure.

Confidence (1-5) and disconfirming evidence

Confidence: 5/5.

Disconfirming evidence to watch: - Evidence that AI efficiency gains materially lower MW demand growth faster than infrastructure can respond. - Rapid, sustained normalization in transformer/interconnection lead times (would reduce scarcity premium). - A policy regime that compresses power-asset returns despite rising demand (for example aggressive price intervention).

Q8) C9: Should We Invest in Department of Defence and War Stocks?

Question

Should we invest in Department of Defence and war stocks, especially with an AGI/ASI arms race between nations in mind?

Prior answer (Mar 31 baseline)

- The prior conclusion was "yes, but selectively."
- It favored asymmetric AI-native defense names (`AVAV`, `KTOS`) over legacy mega-primes.
- It argued that traditional primes are broad defense conglomerates with lower direct AI-bottleneck leverage and are already captured in the Continuity Fund.

What changed since Mar 31

- The core thesis is unchanged: AI-enabled autonomy, affordable mass systems, and software-defined battlefield operations continue to accelerate.
- The investable edge remains in companies tied to autonomy and scalable systems, not in broad platform incumbents with diluted exposure.
- Current V9.0 already keeps defense sizing disciplined (2% total), which still fits risk-adjusted portfolio design.

Updated answer (May 10)

Short answer: **Yes, but as a targeted satellite exposure, not a major core sleeve.**

Defense is strategically relevant to the AGI race, but it is not the primary monetization bottleneck for your portfolio objective (which remains physical AI infrastructure). It should stay as an asymmetric optionality sleeve.

Why defense exposure is rational in an AGI race

1. **AGI competition is geopolitical, not purely commercial.** - Frontier AI leadership has military, intelligence, cyber, and industrial implications. - Public procurement and national-security priorities can create durable demand for selected AI-adjacent defense systems.

2. **Procurement doctrine is shifting toward "affordable mass."** - More autonomous, lower-cost, higher-volume systems are becoming strategically favored. - This tends to benefit specialized autonomy platforms more than legacy exquisite-platform programs.
3. **AI-native defense software and autonomy stack are rising in value.** - ISR automation, sensor fusion, mission autonomy, and human-machine teaming become central. - Companies aligned with autonomous systems and rapid iteration cycles can compound faster than hardware-heavy incumbents.

Why defense should remain a limited sleeve in this portfolio

1. **Not the primary EHF bottleneck layer.** - Defense exposure can contribute alpha, but it does not dominate the same unavoidable throughput stack as power/foundry/memory/grid.
2. **Program and budget volatility.** - Contract timing, appropriations cycles, and program cancellations create episodic risk. - Even good technology can underperform if procurement cadence slips.
3. **Binary event risk at small-cap/ mid-cap defense names.** - Upside is asymmetric, but drawdowns can be sharp around contract outcomes.
4. **Large primes already indirectly owned.** - Broad defense beta is already present in the Continuity Fund via market-cap index exposure.

Investable landscape: which defense names fit this thesis?

Segment	Better-aligned names	Why they fit	Why not bigger weight
Autonomous tactical systems	`AVAV`, `KTOS`	Directly tied to affordable-mass autonomy doctrine; high operational AI relevance	Higher volatility and contract concentration risk
AI-enabled defense software / command stack	`PLTR` (already in ops/control sleeve)	Data fusion, command decision support, deployment adjacency	Multi-sector business mix; not pure defense
Large defense primes	`LMT`, `RTX`, `NOC`, `GD`	Stable backlog and scale	Lower direct AI-bottleneck purity; broad conglomerate exposure
Defense ETFs / broad baskets	`ITA`, etc.	Diversified defense exposure	Dilutes EHF bottleneck focus; overlap with Continuity Fund

Updated assessment of current holdings

`AVAV` (1%) - Role: asymmetry on autonomous systems adoption and field deployment. - Merit: direct alignment with expendable/autonomous tactical demand. - Risk: program concentration and execution volatility.

`KTOS` (1%) - Role: affordable autonomy + collaborative combat aircraft and related systems. - Merit: strong thematic fit for AI-enabled defense doctrine. - Risk: award timing and milestone execution.

Combined, these two names preserve upside to defense-autonomy acceleration without over-allocating capital away from core infrastructure bottlenecks.

Should we add large defense primes?

Current answer: **not required for EHF.**

Rationale: - They are quality companies, but AI-linked upside is diluted by non-AI program mix. - They do not own the same scarcity economics as your primary bottleneck holdings. - Their inclusion would likely lower thematic intensity rather than improve risk-adjusted thesis capture inside EHF.

Should defense sleeve be increased above 2%?

Base-case recommendation: **keep at 2% for now.**

Conditions that could justify a future increase to 3%: - Repeated, high-confidence contract validations in autonomy-focused programs. - Evidence that affordable-mass procurement is scaling faster than currently priced. - No deterioration in execution quality or balance sheet risk for existing holdings.

If increased, it should be funded from lower-conviction non-bottleneck exposures, not from core power/compute bottleneck sleeves.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Current structure (targeted defense asymmetry via `AVAV` + `KTOS`, with `PLTR` as adjacent control/software exposure) remains appropriate.
- Keep defense as a tactical optionality sleeve, not a primary core sleeve.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - A procurement regime shift back toward slow-cycle, legacy platform spending at the expense of autonomy. - Repeated program delays/cancellations in affordable-mass/autonomous initiatives. - Better risk-adjusted AI-defense exposure emerging outside current holdings (for example, a clearly superior public pure-play).

Q9) C12: The New Oil - Can We Invest in Proprietary Data as a Bottleneck?

Question

Can we invest in proprietary data as a bottleneck?

Prior answer (Mar 31 baseline)

- The prior answer concluded that proprietary data is a real moat but difficult to access as a pure-play public bottleneck.
- It argued that most leading data assets sit inside hyperscalers/platforms already captured via the Continuity Fund.

- It also noted that the cleanest EHF-adjacent expression of "data operations" is `PLTR`, not a dedicated data-licensing pure-play.

What changed since Mar 31

- The strategic importance of proprietary data remains high, but the public investable set is still fragmented and mostly indirect.
- Leading data-moat winners continue to be broad platforms, index-heavy incumbents, or private companies rather than focused public bottleneck names.
- The key update for portfolio construction is unchanged: data is often a **software/application moat**, while EHF is optimized for **physical throughput bottlenecks**.

Updated answer (May 10)

Short answer: **yes, proprietary data can be a moat; no, it is usually not an investable bottleneck in the same way as power/foundry/memory.**

To qualify as an EHF-grade bottleneck, a candidate should pass four tests: 1. Non-substitutable input to AI capability deployment. 2. Oligopolistic supply structure with pricing power. 3. High switching cost and low commoditization risk. 4. Public-market vehicle with clean thematic exposure.

Most "data moat" names fail one or more tests for this portfolio framework.

Why proprietary data matters (and where it does create value)

- Proprietary data can improve model quality, relevance, and defensibility in narrow verticals.
- It can create durable feedback loops (better product -> more users -> better data -> better model).
- In enterprise and regulated sectors, curated historical datasets can form meaningful compliance and trust moats.

These are real economic advantages, but they are often embedded within diversified businesses rather than separable bottleneck assets.

Why data usually fails the EHF bottleneck test

1. **Data is often contextual, not universally scarce.** - Many datasets are valuable in-domain but not globally irreplaceable like EUV tools or grid interconnection rights.
2. **Monetization is often bundled.** - Data value is frequently captured through software products or platform bundles, not standalone pricing power on raw data access.
3. **Replication pressure can be high.** - Synthetic data, open-source corpora, and weakly supervised pipelines can erode portions of proprietary advantage over time.
4. **Best-in-class data assets are often inside mega-caps.** - That makes them better suited to Continuity Fund capture than concentrated EHF sleeves.

Investable landscape (public + private context)

Candidate type	Examples	Strength	Limitation for EHF
Hyperscaler/platform data moats	`GOOGL`, `META`, `MSFT`, `AMZN`	Massive proprietary interaction data loops	Buyers of infrastructure; already captured in Continuity Fund
Financial/professional data incumbents	`SPGI`, `MCO`, `FICO`, `LSEG`	Sticky regulated datasets + workflow lock-in	High quality, but not core AI infrastructure bottlenecks
AI data-ops / ontology platforms	`PLTR`	Strong data integration, operational deployment relevance	Multi-factor business; not pure proprietary-data licensing play
Private labeling / RLHF data firms	Private players (e.g., Scale-like category)	Strategic role in training pipelines	Not publicly investable (or too early/speculative)
Niche public "data story" names	Various small caps	Potential upside	Fragile moats, liquidity/quality risk, and weak bottleneck purity

Cross-check against current portfolio

- `PLTR` already captures the most practical, investable edge in this theme: turning fragmented enterprise/government data into operational AI decision systems.
- `CRWD` and `SNPS` also carry indirect data-moat characteristics through telemetry/intelligence and design-flow datasets, but their core ownership thesis in EHF is still security/EDA bottlenecks.
- Adding a separate "data moat" sleeve right now likely increases overlap and complexity without improving bottleneck purity.

What would make a proprietary-data stock add-worthy in future?

A candidate becomes add-worthy (1-2% starter sleeve) only if: - It is publicly listed with sufficient scale/liquidity. - It controls a hard-to-replicate dataset essential to high-value AI workflows. - Revenue model shows direct pricing power from data access/use, not just bundled software upsell. - Evidence of durable retention/switching cost in mission-critical deployments. - Valuation does not require perfection to justify inclusion.

Bottom line for this question

Proprietary data is strategically important, but for this portfolio architecture it is mostly a second-order moat layered on top of software/platform business models. The Event Horizon Fund is currently better optimized by keeping the focus on physically binding infrastructure bottlenecks and using existing `PLTR` exposure for data-operations adjacency.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Do not add a dedicated proprietary-data position at this time.
- Keep `PLTR` as the practical data-ops/control expression and revisit only if a clear public pure-play data bottleneck emerges.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - A public company emerges with genuinely non-substitutable AI training/operational data and clear standalone pricing power. - Regulatory/data-rights shifts materially increase the scarcity value of certain proprietary datasets. - Evidence that data-control moats are outperforming physical bottlenecks on a sustained multi-year basis in the AI value chain.

Q10) C2: Should I Invest in Identity / Provenance / Compliance Rails?

Question

Should I invest in identity, provenance, or compliance rails?

Prior answer (Mar 31 baseline)

- The prior conclusion was "not yet, but monitor closely."
- It identified the problem as real and growing (deepfakes, synthetic identity fraud, AI-generated content authenticity), but investable public pure-plays were limited.
- It argued that current exposure is partially captured through existing holdings (`CRWD`, `PLTR`) rather than a dedicated new sleeve.

What changed since Mar 31

- The threat trajectory remains intact: authenticity, attribution, and policy-compliant AI deployment continue to become more important.
- Market structure still looks early and fragmented; standards are progressing but monetization is uneven.
- The best public names remain multi-product/platform companies, not narrow pure-play "identity/provenance rail" monopolies.

Updated answer (May 10)

Short answer: **the theme is real and likely huge over time, but still early for a dedicated EHF sleeve unless a clearer public winner emerges.**

Identity/provenance/compliance rails are likely to become foundational as AI-generated content and autonomous agents scale. But today, the investable universe is mostly "adjacent exposure," not bottleneck purity.

Why this category matters in an AGI/ASI trajectory

As model capability rises, three control problems compound: 1. **Who is acting?** (identity and authorization for agents, users, systems) 2. **What is authentic?** (content provenance, watermarking, chain-of-custody) 3. **What is permitted?** (policy, auditability, and regulatory compliance controls)

Without robust rails, fraud, legal uncertainty, and operational risk rise fast enough to slow enterprise and government deployment.

Threat and demand drivers (structural, not temporary)

- Synthetic identity fraud and deepfake-enabled social engineering are rising in sophistication.
- Enterprises need machine-verifiable provenance for legal, financial, and regulated workflows.
- Governments are moving toward stronger AI accountability requirements (model governance, audit trails, and risk controls).
- As autonomous agents gain permissions, identity and policy enforcement move from "security feature" to mission-critical control plane.

Bottleneck test: does this qualify as an EHF-style bottleneck today?

Test	Status today	Interpretation
Non-substitutable input to AI deployment	**Partially yes**	Required eventually, but multiple overlapping approaches still exist
Oligopolistic supplier structure	**No (yet)**	Fragmented vendor landscape; no clear durable monopoly
Strong standalone pricing power	**Mixed**	Often bundled into broader security/workflow platforms
Clean public pure-play	**Weak**	Most candidates are private, micro-cap, or diversified

Net: strategically important, but currently below the threshold for a dedicated concentrated sleeve in this portfolio.

Investable landscape (current state)

Segment	Representative names	Why relevant	Limitation for EHF now
Identity/security platforms	`CRWD`, `OKTA`, large IAM ecosystems	Directly address identity attacks, auth controls, policy enforcement	Crowded category; value often bundled; no clear single bottleneck owner
Enterprise/government control software	`PLTR`	Strong governance, workflow control, and auditability in mission contexts	Multi-product exposure; not a pure provenance rail
Public safety/ compliance-adjacent	`AXON`	Chain-of-evidence and digital trust in law-enforcement workflows	Vertical-specific application, not broad AI control-rail bottleneck
Content provenance standards ecosystems	C2PA-linked corporate participants	Potential long-run standard layer for authenticity	Standards still monetization-early and diffuse
Private deepfake detection / identity startups	Private leaders in liveness/provenance	Potentially high strategic value	Not public or too early/ speculative for concentrated sleeve

How current portfolio already captures the theme indirectly

- **`CRWD`** provides practical identity-threat and AI-security exposure in deployed enterprise environments.

- `PLTR` provides control-plane and governance exposure in operational enterprise/government contexts.
- Broad platform exposure in the Continuity Fund captures additional identity/provenance R&D indirectly.

This means you are not unexposed; you are just avoiding premature overconcentration in a market that has not yet consolidated.

What would justify adding a dedicated identity/provenance/compliance rail position?

Add only if most of these conditions are met: - Public company with sufficient scale/liquidity and durable growth quality. - Evidence of platform-level lock-in (high net retention, low churn, mission-critical integration). - Regulatory tailwind translates into recurring, non-discretionary spend. - Product acts as a true gatekeeper layer (identity/provenance checks required for downstream workflows). - Valuation does not assume flawless execution.

Potential sizing if criteria are met: **1-2% starter allocation**, funded from lower-conviction non-bottleneck exposures.

Why this is a "monitor aggressively" theme rather than "buy now" theme

This category can become a major winner later, but timing matters: - Too early: high narrative risk, weak pricing power durability. - Too late: you miss lock-in compounding.

Current evidence favors a watchlist posture with explicit triggers, while using existing `CRWD` + `PLTR` exposure as pragmatic interim capture.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Do not add a dedicated identity/provenance/compliance rail position yet.
- Maintain current indirect exposure (`CRWD`, `PLTR`) and revisit when market structure and monetization become more concentrated/clear.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - One public platform becomes a de facto required identity/provenance gate across major AI workflows. - Regulation rapidly creates non-discretionary spend concentrated in a small number of providers. - Clear, sustained evidence that identity/provenance rails are capturing more economic value than expected from current adjacency exposures.

Q11) C1: Should I Invest in Domestic Surveillance & Public Safety?

Question

Should I invest in domestic surveillance and public safety?

Prior answer (Mar 31 baseline)

- The prior answer identified `AXON` as the strongest public-market company in this category but concluded it was not a core EHF bottleneck fit.
- It distinguished "excellent company" from "best thesis alignment" and favored keeping EHF concentrated on infrastructure bottlenecks.
- It noted this category is often better captured via broad index exposure unless a clear bottleneck owner emerges.

What changed since Mar 31

- The strategic relevance of surveillance/public-safety AI increased as real-time video, sensor fusion, and AI-assisted reporting expanded.
- However, market structure is still fragmented across private firms, diversified mega-caps, and niche hardware vendors.
- The core portfolio-construction question is unchanged: does this category offer a scarce, non-substitutable bottleneck with clean public investability?

Updated answer (May 10)

Short answer: **yes, the sector is important and likely to grow, but mostly as an application/control layer rather than a primary infrastructure bottleneck for EHF.**

A targeted, limited exposure can make sense only when the specific company has durable lock-in and clear pricing power. Broadly adding "surveillance stocks" as a sleeve would likely dilute thesis purity.

Why the theme matters

1. **AI sharply increases surveillance system capability.** - Real-time detection, anomaly spotting, object/person re-identification, and automated summarization raise operational value.
2. **Public safety workflows are moving from hardware to software + recurring cloud evidence systems.** - The moat increasingly sits in data workflow lock-in, chain-of-custody, and operational software, not only devices.
3. **Government and enterprise adoption can be sticky once integrated.** - Procurement cycles are slow, but once deployed, switching costs can be meaningful due to retraining, policy workflows, and evidence-process integration.

Bottleneck test for this category in EHF context

Test	Result	Practical interpretation
Non-substitutable physical bottleneck	Mostly **No**	More of an application/platform layer than a hard physical choke point
Oligopoly/monopoly structure	**Mixed**	Some local dominance pockets, but broad market still competitive
Durable pricing power	**Mixed**	Stronger in software workflows than in camera hardware
Clean public pure-play access	**Partial**	Best candidates are either private or embedded in diversified companies

Net: category is investable in principle, but only selectively and usually at modest size for this portfolio style.

Company-by-company investigation (including requested names)

Company / Ecosystem	Public?	What it does	Thesis fit for EHF	Verdict
Flock Safety	No (private)	ALPR + neighborhood/ public safety camera network + software workflows	Strong category signal; recurring software + network effects	**Monitor only** (not publicly investable)
Ring (Amazon)	No standalone (`AMZN`)	Consumer doorbell/camera ecosystem	Massive installed base but bundled inside hyperscaler/ retail giant	**No direct add** (captured via Continuity Fund)
Google Nest (Alphabet)	No standalone (`GOOGL`)	Consumer/home camera and smart-home surveillance stack	Similar to Ring: real footprint, but buried in diversified mega-cap	**No direct add** (captured via Continuity Fund)
Arlo (`ARLO`)	Yes	Consumer/security cameras + subscription services	Public pure-play exposure exists, but weaker moat and higher competitive pressure	**Watchlist only** (not strong enough for core EHF sleeve)
Meta smart glasses / wearable capture (`META`)	No standalone	Wearable cameras and ambient-compute interface trend	Potentially important long-term input layer, but embedded in mega-cap platform strategy	**No direct add** (captured via Continuity Fund)
Axon (`AXON`)	Yes	Body cameras, digital evidence platform, AI-assisted report/public safety workflow	Best public "public safety + software lock-in" expression	**Highest-quality candidate in category; still optional for EHF**
Palantir (`PLTR`)	Yes	Data integration and operational software in gov/ public safety contexts	Already held; stronger control-plane fit than hardware-centric surveillance	**Already captured**

Detailed view: `AXON` vs requested alternatives

- **Vs private network models (`Flock`)**: `AXON` is public and has deep workflow lock-in; `Flock` may be strong but cannot be owned directly today.
- **Vs consumer camera ecosystems (Ring/Nest/Arlo)**: consumer hardware has faster commoditization risk and weaker moat durability than mission-critical evidence workflows.
- **Vs wearable camera trend (Meta glasses)**: strategically interesting, but monetization is currently platform/ecosystem-driven rather than a clean public-safety bottleneck.

Conclusion: if you want one public name in this category, `AXON` is typically the strongest quality expression. The question is not quality; it is sleeve fit.

Should EHF add `AXON` now?

Base-case: **not required right now** for this portfolio.

Why: - EHF is optimized for hard AI infrastructure bottlenecks (power, foundry, memory, optical, cooling). - `AXON` is more of an application/workflow moat in a specific vertical. - Adding it would increase thematic breadth but reduce bottleneck concentration.

When it could become add-worthy (1-2% starter): - Sustained proof that evidence/command software becomes mandatory infrastructure layer across agencies. - Continued high-quality recurring revenue growth with durable retention and expanding software mix. - Funding source comes from lower-conviction non-bottleneck positions, not from core power/compute chokepoints.

Practical portfolio positioning

- Current setup already has partial exposure through:
- `PLTR` (operations/control in government/enterprise contexts)
- Continuity Fund holdings in `AMZN` / `GOOGL` / `META` (indirect surveillance ecosystem exposure)
- This reduces urgency to force a dedicated surveillance sleeve inside EHF.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Keep domestic surveillance/public safety as a monitored adjacency theme, not a primary EHF bottleneck sleeve.
- If you want discretionary optionality, `AXON` is the only high-quality public candidate worth serious consideration, and even then at small size (1-2% max) with explicit criteria.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - A public safety platform becomes a de facto required operating layer with clear monopoly-like economics. - Regulatory mandates create non-discretionary nationwide spending with concentrated vendor winners. - Consumer/wearable surveillance ecosystems unexpectedly consolidate into a dominant, high-margin, hard-to-replicate public pure-play.

Q12) C7: Should We Invest When Bio-Engineered Pandemics Become Easily Accessible?

Question

Should we invest in stocks like `MRNA` when bio-engineered pandemics as bad or worse than COVID become increasingly accessible via open-source or jailbroken models?

Prior answer (Mar 31 baseline)

- The prior answer treated this as a **tail-risk hedge sleeve**, not a core return engine.
- It favored maintaining `MRNA` as asymmetric "insurance" exposure rather than expanding into a broad biotech bet.

- It emphasized that this sleeve should stay small and purpose-built because timing is uncertain and carry costs can be high.

What changed since Mar 31

- The core logic is unchanged: bio-risk remains low-frequency but potentially extreme-severity.
- AI-enabled capability diffusion keeps pressure on preparedness frameworks and response-speed infrastructure.
- For portfolio design, the key remains asymmetry: small allocation, large potential payoff under adverse tail scenarios.

Updated answer (May 10)

Short answer: **Yes, but as a small, explicit catastrophe-hedge sleeve, not as a growth sleeve.**

This is one of the few areas where expected value is driven less by base-case earnings compounding and more by convex payoff under systemic shock.

Why this risk is financially relevant

1. **Tail severity is enormous.** - A severe engineered-pandemic scenario can produce simultaneous human, economic, and policy shocks. - Under such conditions, rapid-response platform companies can re-rate sharply.
2. **Preparedness value is accelerating.** - Speed-to-design, speed-to-manufacture, and speed-to-distribution become core strategic assets. - mRNA platform flexibility remains one of the strongest known public-market expressions of rapid countermeasure deployment.
3. **Market underweights low-frequency risks in calm periods.** - This can make small, disciplined hedge exposure attractive before crisis pricing appears.

Why this should remain small in EHF

1. **Timing uncertainty is extreme.** - Great hedge logic does not imply good short-term compounding.
2. **Opportunity cost exists.** - Capital allocated here is capital not allocated to first-order AI infrastructure bottlenecks.
3. **Biotech execution/clinical risk is real.** - Even strong platforms can face long periods of valuation compression outside crisis windows.

So the right construction is not "bet big on pandemics"; it is "hold a right-sized asymmetric hedge."

`MRNA` fit assessment (why it remains the chosen vehicle)

Criterion	`MRNA` assessment	Why it matters
Platform speed and adaptability	Strong	Core to rapid-response thesis in high-uncertainty pathogen scenarios
Public liquidity and scalability	Strong	Practical for maintaining a hedge position over time

Criterion	`MRNA` assessment	Why it matters
Directness to bio-risk hedge thesis	High	Cleaner expression than broad biotech baskets
Carry profile outside crisis windows	Mixed	Can underperform in non-event periods

Net: `MRNA` still fits the role best as an **asymmetric, explicit hedge** at small size.

Alternatives considered

Alternative	Strength	Why not superior for this specific sleeve
`BNTX`	High scientific quality and mRNA relevance	Comparable but does not clearly dominate as a hedge vehicle in this portfolio
`PFE`	Global scale, manufacturing/distribution reach, deep balance-sheet resources	More diversified pharma profile dilutes pure asymmetric bio-risk convexity; less direct "tail-hedge torque" than a focused platform expression
Broad biotech ETF (`IBB`, `XBI`)	Diversification	Dilutes catastrophe-hedge asymmetry; becomes general biotech beta
Diagnostics/public-health services names	Could benefit in some outbreaks	Less direct convexity to rapid vaccine/countermeasure platform dynamics
No allocation	Avoids carry drag	Leaves portfolio unhedged to an extreme but plausible systemic risk

How to think about expected value

This sleeve is not optimized for average-quarter alpha.
It is optimized for **portfolio resilience under extreme downside scenarios** where: - broad risk assets sell off, - policy urgency accelerates countermeasure demand, - rapid-response bioplatforms can experience sharp relative outperformance.

In that framework, a small allocation can improve total portfolio robustness even if standalone long periods are choppy.

Sizing and discipline framework

Current sizing logic: keep bio-risk hedge in a **small band (about 1%)**.

Increase consideration (for example toward 1.5-2%) only if: - clear deterioration in global biosecurity risk profile, - policy/funding regime materially increases preparedness spending durability, - and funding source is lower-conviction non-bottleneck exposure (not core infrastructure chokepoints).

Reduce/exit consideration if: - better public hedge vehicle emerges with superior platform resilience and valuation, - or thesis degrades due to persistent platform competitiveness loss.

Portfolio implication

- **No immediate allocation change required from this question alone.**

- Keep `MRNA` as a small, explicit asymmetric bio-risk hedge sleeve.
- Do not replace with broad biotech exposure inside EHF; that is better handled by the Continuity Fund.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - Durable evidence that `MRNA` loses relative platform competitiveness in rapid-response countermeasure development. - A clearly superior public alternative emerges for catastrophe-hedge convexity. - Structural decline in bio-risk relevance due to credible global prevention architectures that materially lower severe-scenario probability.

Q13) C8: Should We Invest in Anonymity, Privacy, VPNs, Crypto, Blockchain?

Question

Should we invest in anonymity and/or anonymity-unmasking themes (VPNs, currencies, wallets, crypto, blockchain, and online-presence infrastructure)?

Prior answer (Mar 31 baseline)

- The prior logic treated this as a real societal and security trend but weakly aligned with EHF's physical bottleneck framework.
- It suggested selective monitoring rather than immediate dedicated allocation.
- It emphasized that many obvious candidates are either private, speculative, or structurally outside the portfolio's core thesis.

What changed since Mar 31

- The strategic relevance of digital identity, anonymity, and anti-anonymity tooling increased alongside agentic AI, deepfakes, and fraud automation.
- However, public investability remains fragmented: many strongest products sit inside diversified platforms or private companies.
- The key construction question is still whether these exposures behave like durable bottlenecks or like high-volatility application/speculation trades.

Updated answer (May 10)

Short answer: **mostly no for a dedicated core EHF sleeve right now; monitor selectively where it overlaps with security/control rails already in portfolio.**

This space is important, but most of it does not currently meet EHF's bottleneck criteria (scarcity, replacement difficulty, durable pricing power, clean public expression).

First principles: separate the theme into distinct buckets

Bucket	What it includes	EHF bottleneck fit today	Why
Privacy / anonymity tools	VPNs, privacy browsers, encrypted comms ecosystems	Low-to-moderate	Real demand, but limited durable public pure-play moats; often commoditizing
Identity unmasking / trust enforcement	Fraud detection, identity verification, provenance/ authenticity rails	Moderate (rising)	Strategically important, but market still fragmented and bundled
Crypto / wallets / on-chain rails	Exchanges, custodians, wallets, base-layer token ecosystems	Low for EHF core	High cyclicality/regulatory sensitivity; weaker direct tie to physical AI bottlenecks
Online-presence graph/control	Platform identity ecosystems, device-linked presence layers	Mixed	Strong strategic value but mostly embedded in mega-cap platforms

Why this category is difficult to allocate in EHF

1. **Many candidates are applications, not bottlenecks.** - They can grow fast but often do not own unavoidable physical or regulatory chokepoints.
2. **High regime dependence.** - Regulatory cycles can rapidly re-rate privacy and crypto businesses.
3. **Public-market purity problem.** - Strongest ecosystems are frequently inside diversified giants or private companies.
4. **Overlap with existing sleeves.** - Portions of "anonymity unmasking" are already covered through security/control holdings.

Investable landscape (public lens)

Segment	Representative exposures	Strength	Why not superior for EHF now
Cybersecurity identity/fraud controls	`CRWD` (held), adjacent IAM names	Directly tied to real enterprise risk spend	Already captured; category still broad and competitive
Ops/control governance layers	`PLTR` (held)	Strong in high-trust workflows and gov/enterprise control contexts	Already captured; not a pure anonymity rail
VPN/privacy software vendors	Public and private niche providers	Demand exists and may rise with AI-enabled fraud/content concerns	Moats are uneven; many products face pricing pressure/ commoditization
Crypto exchanges / wallets / miners	Various public crypto-linked equities	High upside in specific cycles	Cyclical beta and policy risk; weak tie to AI infrastructure bottlenecks
Big-platform identity ecosystems	`META`, `GOOGL`, `AAPL`, `MSFT`, etc.	Massive scale and data advantages	Already captured in Continuity Fund; adding directly in EHF dilutes focus

Requested sub-topics: direct answers

VPNs / privacy tools:
Investable as a tactical theme, but generally not a top-tier EHF bottleneck. Better as watchlist unless a clear winner shows durable enterprise-grade lock-in and pricing power.

Currencies / wallets / crypto / blockchain:
Potentially high-upside in cycle terms, but currently more macro/regulatory/speculative than bottleneck-pure for this strategy. Not preferred as a core EHF sleeve at this time.

Anonymity unmaskers (identity/provenance/fraud rails):
Most relevant subtheme for EHF, but better captured for now by current holdings (`CRWD`, `PLTR`) while market structure matures.

What would make this theme add-worthy later?

A dedicated 1-2% sleeve could be justified only if: - A public company emerges as a de facto required trust/identity/provenance gate. - Revenue becomes non-discretionary and deeply embedded across enterprise/government workflows. - Competitive position shows durable moat (retention, switching costs, ecosystem dependence). - Valuation supports risk-adjusted upside without requiring perfect execution.

Risk of adding too early

- You may pay growth multiples before moat durability is proven.
- Thematic overlap can creep in and dilute concentration in the highest-confidence physical bottlenecks.
- Volatility can increase without improving long-horizon thesis capture.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Do not add dedicated VPN/crypto/privacy sleeve to EHF right now.
- Keep exposure focused on existing trust-enforcement adjacency (`CRWD`, `PLTR`) and reassess if a true public bottleneck winner emerges.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - A single public trust/identity platform becomes mandatory infrastructure across major AI transaction/workflow layers. - Regulation creates concentrated winner-take-most economics in provenance or identity enforcement. - Clear sustained outperformance of trust-rail businesses with durable moats versus current adjacency exposures.

Q14) Early Thesis Invalidation: What Could Prove the Bottleneck Thesis Wrong by 2027?

Question

What five developments, ranked, would most strongly invalidate the Event Horizon Fund bottleneck thesis by 2027?

Prior answer (Mar 31 baseline)

- Prior sections had company-level falsification triggers, but there was no single ranked "portfolio-level invalidation list."
- The architecture implicitly assumed AI demand growth would remain constrained by physical throughput bottlenecks.
- The key gap was explicit ranking of what could break that assumption early.

What changed since Mar 31

- The thesis is still validating in current data, but concentration risk makes explicit invalidation criteria more important, not less.
- With added sleeves (`BE`, `COHR`) and elevated power-chain weight, a portfolio-level kill-switch framework is now essential risk discipline.
- By 2027, we should expect enough evidence to materially confirm or falsify the bottleneck-first framework.

Updated answer (May 10)

Short answer: the thesis is invalidated early only if the economy's binding constraint shifts away from scarce physical infrastructure faster than expected.

Below are the **top 5 invalidation developments, ranked by impact**.

Ranked invalidation list (strongest first)

Rank	Development that would invalidate thesis	Why this is high-impact	Early evidence to watch by 2027
1	**Physical bottlenecks de-bottleneck quickly and broadly** (power interconnection, transformers, packaging, optical)	The thesis depends on scarcity pricing power. If constraints clear quickly, bottleneck premiums compress.	Sustained lead-time normalization, falling backlog pressure, lower pricing power across power/electrical/infra suppliers.
2	**Algorithmic efficiency shock sharply reduces infrastructure intensity** (much less MW/FLOP and hardware per unit capability)	If capability growth decouples from physical buildout, downstream software captures more value than infrastructure owners.	Frontier performance rises while datacenter capex growth and power intensity stall materially.
3	**Hyperscaler capex regime shifts from buildout to optimization**	The portfolio assumes continued heavy physical deployment. A capex pivot undermines multiple sleeves simultaneously.	Multi-quarter cuts in AI infrastructure capex guidance, slower order intake for equipment/power/cooling/optical names.

Rank	Development that would invalidate thesis	Why this is high-impact	Early evidence to watch by 2027
4	**Policy/regulatory intervention caps returns for bottleneck owners**	Even true bottlenecks can become poor investments if returns are administratively compressed.	Price controls, forced contracting terms, mandated pass-through economics, adverse PPA/ regulatory changes.
5	**Geopolitical/industrial realignment reduces U.S.-aligned scarcity premium**	The thesis embeds geopolitical fragmentation and strategic scarcity. Rapid de-risking/capacity coordination would compress premiums.	Durable de-escalation, coordinated supply expansion, lower strategic-premium pricing in key bottleneck lanes.

Why these five matter more than ordinary risks

These are **thesis risks**, not normal volatility risks: - A 20-30% stock drawdown is not invalidation by itself. - Temporary earnings misses are not invalidation by themselves. - Invalidation requires structural change in value-capture mechanics across the AI stack.

Portfolio-level early warning dashboard (2026-2027)

Signal cluster	Bullish for thesis	Bearish for thesis (invalidation direction)
Lead times/ backlogs	Persistent tightness; constrained delivery slots	Rapid normalization across multiple bottlenecks at once
Pricing power	Stable or rising contract economics for scarce capacity	Broad margin compression despite healthy demand
Hyperscaler capex	Continued infrastructure-heavy spend growth	Material pivot to software efficiency with flat/down infra spend
Power system stress	Interconnection constraints and reliability tightness persist	Fast queue clearing + excess reserve buildout
Market leadership	Bottleneck owners continue relative outperformance	Durable shift to downstream app/software names detached from infra buildout

Severity and action mapping

Invalidation level	Description	Recommended portfolio action
Level 0 (None)	Thesis continues validating	Hold core structure; rebalance tactically only
Level 1 (Mild)	One warning cluster weakens	Trim weakest execution names first; keep architecture intact
Level 2 (Moderate)	Two clusters weaken for 2+ quarters	Reduce cyclical bottleneck overweight; raise quality/ defensiveness
Level 3 (Severe)	Three+ clusters weaken structurally	De-risk core thesis concentration; rotate toward broader resilient exposures

What would *not* count as invalidation

- A single geopolitical headline shock without structural demand change.

- One company's execution failure while peers remain strong.
- Short-term valuation compression during risk-off episodes.

Practical interpretation for your current allocation

The current portfolio can tolerate normal volatility, but it should not ignore structural signal shifts. If top-ranked invalidation signals (1-3) appear together, the correct response is **gradual structural rebalance**, not "hold at all costs."

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Add this ranked invalidation framework to ongoing review cadence (quarterly minimum).
- If two or more top-3 invalidation signals persist simultaneously, begin staged de-risking from most valuation-sensitive bottleneck names first.

Confidence (1-5) and disconfirming evidence

Confidence: 5/5 on framework relevance; **4/5** on exact ranking order (order can shift with new data).

Disconfirming evidence to watch: - Continued broad scarcity, strong capex, and resilient pricing power through 2027 would disconfirm invalidation concerns and reaffirm the thesis.

Q15) Architecture Test: Is This Portfolio Optimal vs Alternative Models?

Question

How does this portfolio compare to 2-3 alternative portfolio architectures? Is the current structure truly optimal versus a concentrated-core + tactical-satellite model, or versus a barbell model?

Prior answer (Mar 31 baseline)

- Prior analysis optimized allocations within a single architecture, but did not explicitly compare architecture families side-by-side.
- The implicit model was "integrated bottleneck concentration" with multiple sleeves spanning power, semis, optical, grid, security, defense, and hedge positions.
- The open question remained whether the same thesis could be captured with better risk-adjusted outcomes through a different structure.

What changed since Mar 31

- V9.0 increased architecture complexity (new sleeves like `BE` and `COHR`) while improving bottleneck completeness.
- Concentration is still intentional, but the number of moving parts now makes architecture design itself a meaningful edge variable.
- This is the right time to compare framework-level alternatives, not just ticker-level substitutions.

Updated answer (May 10)

Short answer: **the current integrated bottleneck architecture remains the best fit for your objective**, but a *hybridized version* of concentrated-core + tactical-satellite would likely improve execution discipline without diluting thesis power.

The three architectures compared

Architecture	Definition	Strengths	Weaknesses	Best use case
A) Integrated Bottleneck Portfolio (current V9.0 style)	Broad set of bottleneck sleeves across full AI physical stack	Captures multiple scarcity lanes; less single-point failure; high thematic fidelity	More positions to monitor; risk of over-complexity	High-conviction investor willing to actively review fundamentals
B) Concentrated-Core + Tactical-Satellite	Smaller core in highest-confidence bottlenecks + small tactical sleeves for emerging opportunities	Cleaner risk budget, easier rebalancing, clearer "must-win" exposures	Can miss secondary bottleneck upside if core too narrow	Investor seeking high conviction with tighter operational discipline
C) Barbell Model	Defensive/quality anchor + high-beta optionality tail	Better downside resilience in regime shifts; explicit convexity sleeve	Can dilute main thesis capture if defensive side too broad	Investors prioritizing drawdown control over maximum thematic purity

Architecture A (current): strengths and limits

Strengths - Strong representation across the full deployment chain (power, electrical, foundry, memory, optical, cooling, security/control). - Better resilience if bottleneck location migrates (for example, from compute to power to optical). - Reduces dependence on a single bottleneck narrative.

Limits - Monitoring burden is higher: multiple sleeves need ongoing evidence checks. - More opportunities for "style drift" if additions are made without strict bottleneck filters. - Position sizing discipline becomes the main determinant of risk-adjusted outcomes.

Architecture B: concentrated-core + tactical-satellite (most credible challenger)

How it would look conceptually - **Core (about 70-80% of EHF sleeve):** highest-confidence bottlenecks only (power chain + foundry/tooling + memory + one networking/optical anchor). - **Satellites (about 20-30%):** rotating smaller sleeves (defense asymmetry, bio-risk hedge, emerging trust rails, tactical adds like new bottleneck entrants).

Why this can be superior operationally - Forces hard prioritization. - Improves trim/add discipline (satellites are explicitly tactical by design). - Reduces accidental over-diversification within a concentrated strategy.

Main risk - If core definition is too narrow, portfolio may undercapture secondary bottlenecks that become primary later.

Architecture C: barbell model

How it would look conceptually - One side: defensive/quality compounding exposures. - Other side: high-beta convex plays tied to acceleration scenarios.

Why it is less optimal for your stated objective - Your explicit goal is maximizing upside from AI infrastructure bottlenecks, not balancing toward broad defensiveness inside EHF. - Defensive ballast is already handled at the total-portfolio level through the Continuity Fund.

Result: barbell logic is useful at whole-portfolio level, but less necessary inside the EHF sleeve itself.

Comparative scorecard (for your objective: maximize bottleneck-driven upside with disciplined risk)

Criterion	A) Integrated Bottleneck	B) Core + Satellite	C) Barbell
Bottleneck thesis fidelity	**9/10**	8.5/10	6.5/10
Upside capture potential	**9/10**	8.5/10	7/10
Operational simplicity	7/10	**9/10**	8/10
Drawdown control inside EHF	7/10	8/10	**9/10**
Fit with your current 3-fund setup	**9/10**	9/10	7.5/10
Overall for your stated goal	**8.6/10**	**8.8/10 (slightly better execution profile)**	7.6/10

Recommendation

The best answer is a **hybrid**: - Keep architecture **A's broad bottleneck coverage** (do not abandon the integrated thesis). - Apply architecture **B's governance discipline**: - Label positions explicitly as `Core` vs `Satellite`. - Set tighter max weights and review cadence for satellites. - Use satellite budget as the only place for new thematic experiments.

This keeps upside capture high while reducing complexity creep.

Practical implementation template (no immediate allocation change required)

1. **Tag each holding now as Core or Satellite.**
2. **Set sleeve budget rules** (example: Core floor, Satellite cap).
3. **Require higher evidence threshold** for promoting Satellite -> Core.
4. **Require faster trim discipline** for Satellite names if thesis signals weaken.
5. **Review architecture quarterly**, not only ticker-by-ticker.

Portfolio implication

- **No immediate allocation change required from this question alone.**
- Current V9.0 remains strong, but execution should shift to a clearer **Core + Satellite governance overlay**.
- This is an architecture/process improvement, not a thesis reversal.

Confidence (1-5) and disconfirming evidence

Confidence: 4/5.

Disconfirming evidence to watch: - If bottleneck leadership broadens so much that concentrated governance materially underperforms simpler diversified approaches. - If satellites consistently outperform core with lower risk, indicating core definition is too narrow or stale.